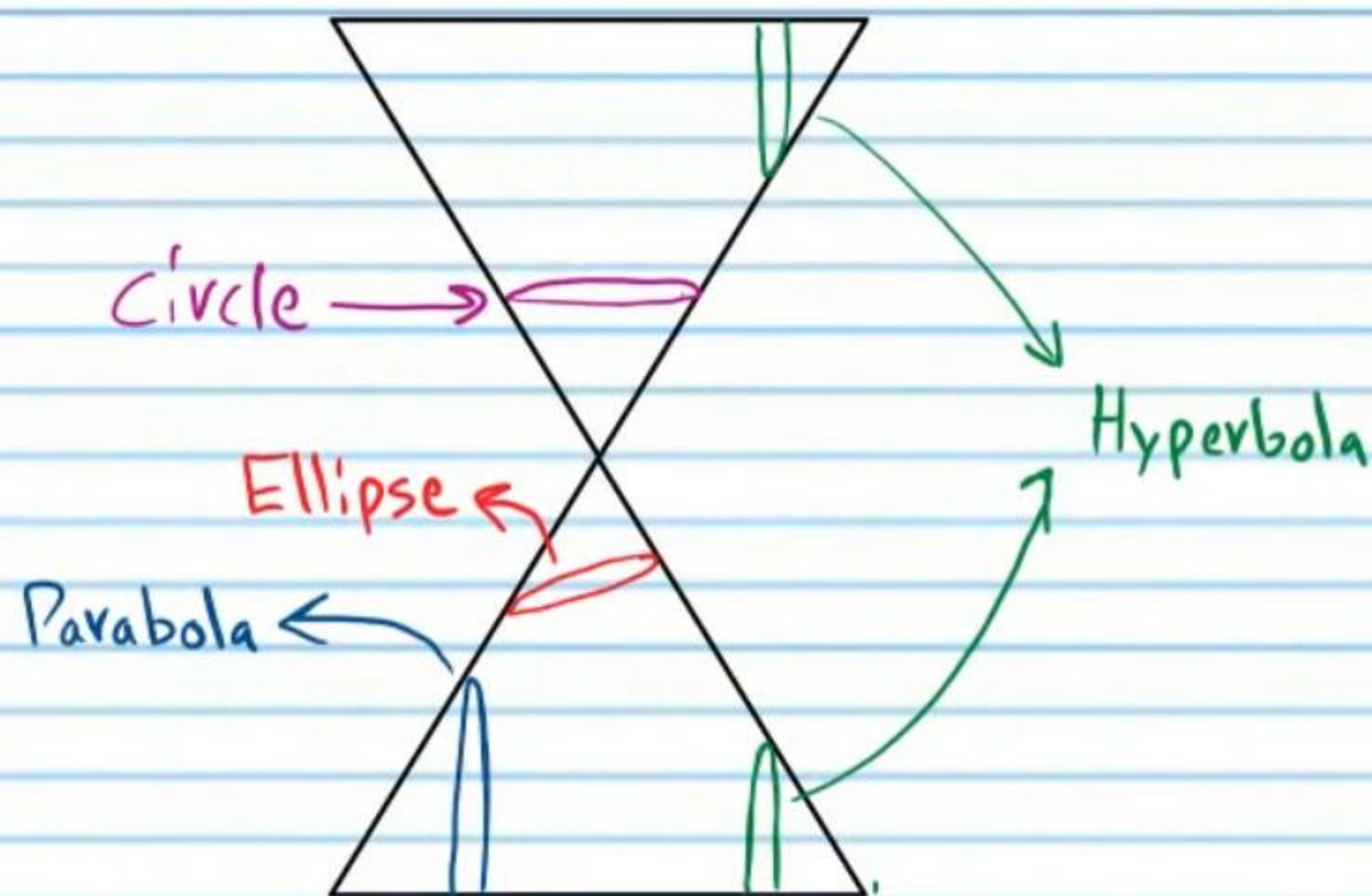


The Conics (Part II)



Equations of the Conics

Parabola

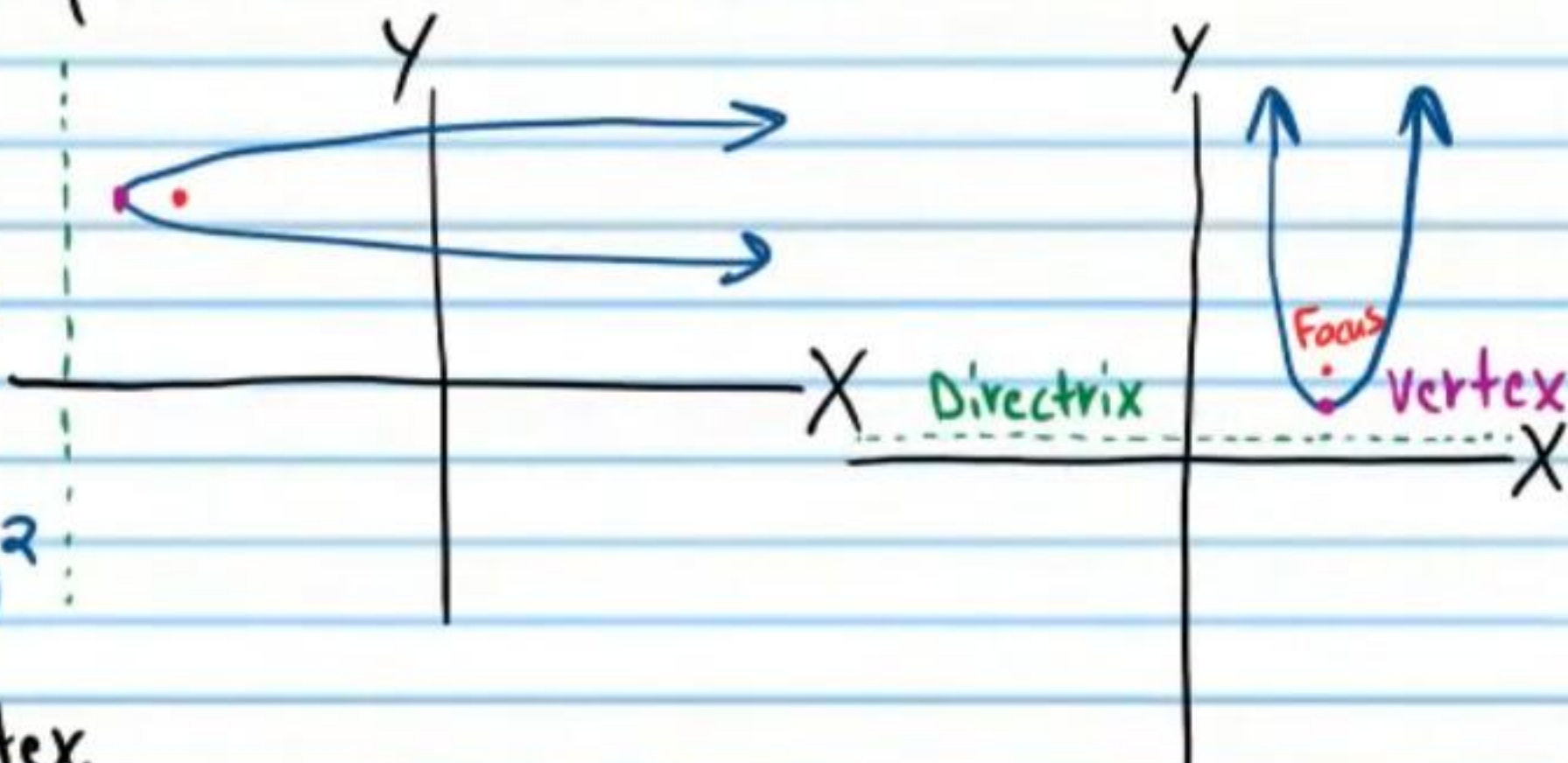
Vertical
 $y - k = a(x - h)^2$

Horizontal
 $x - h = a(y - k)^2$

(h, k) is the vertex.

$a = \frac{1}{4p}$

p is the distance from the vertex to the focus or directrix.



Other Forms

Standard

$y = ax^2 + bx + c$

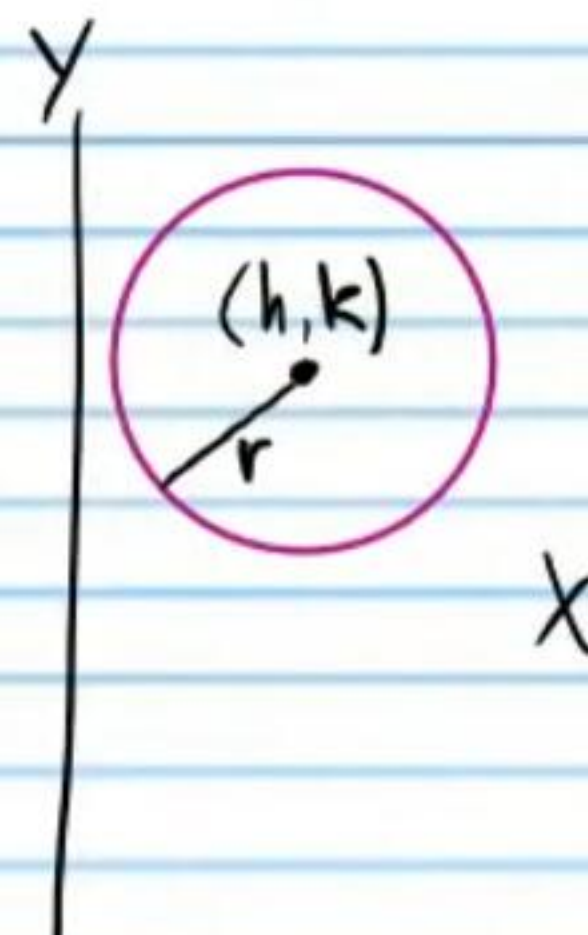
Intercept

$y = a(x - m)(x - n)$

Circle

$(x - h)^2 + (y - k)^2 = r^2$

(h, k) is the center.
 r is the radius.



Ellipse

Horizontal
 $\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1$

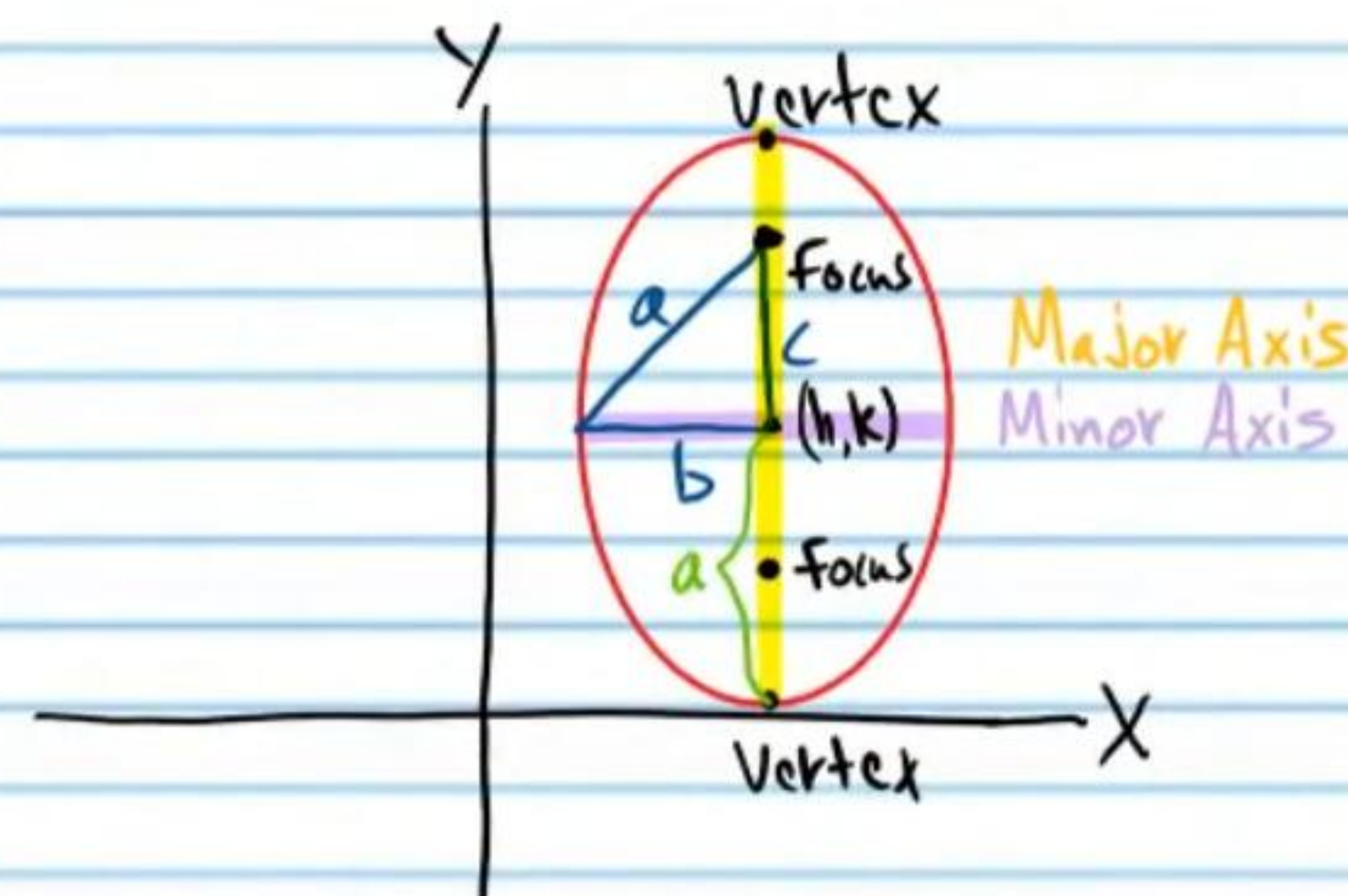
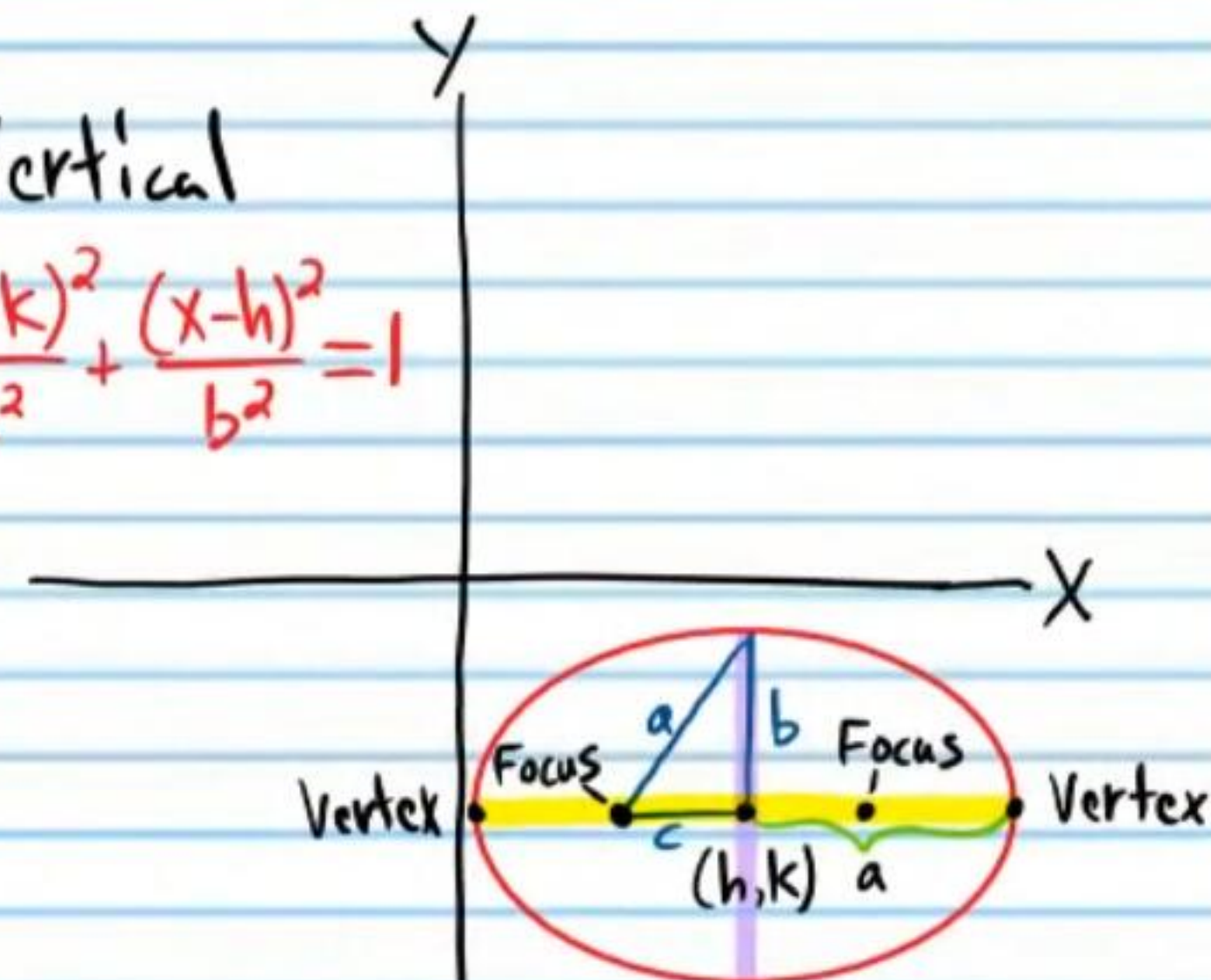
Vertical
 $\frac{(y - k)^2}{a^2} + \frac{(x - h)^2}{b^2} = 1$

Vertical

(h, k) is the center.

$c^2 + b^2 = a^2$

c is the distance along the major axis from the center to the foci.
 a is the distance from the center to the vertices.
 $2a$ is the length of the major axis.
 $2b$ is the length of the minor axis.



Hyperbola

Opens Right and Left

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

Opens Up and Down

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

(h,k) is the center.

$$a^2 + b^2 = c^2$$

c is the distance from the center to the foci. a is the distance from the center to the vertices.

Asymptotes Opening Up/Down

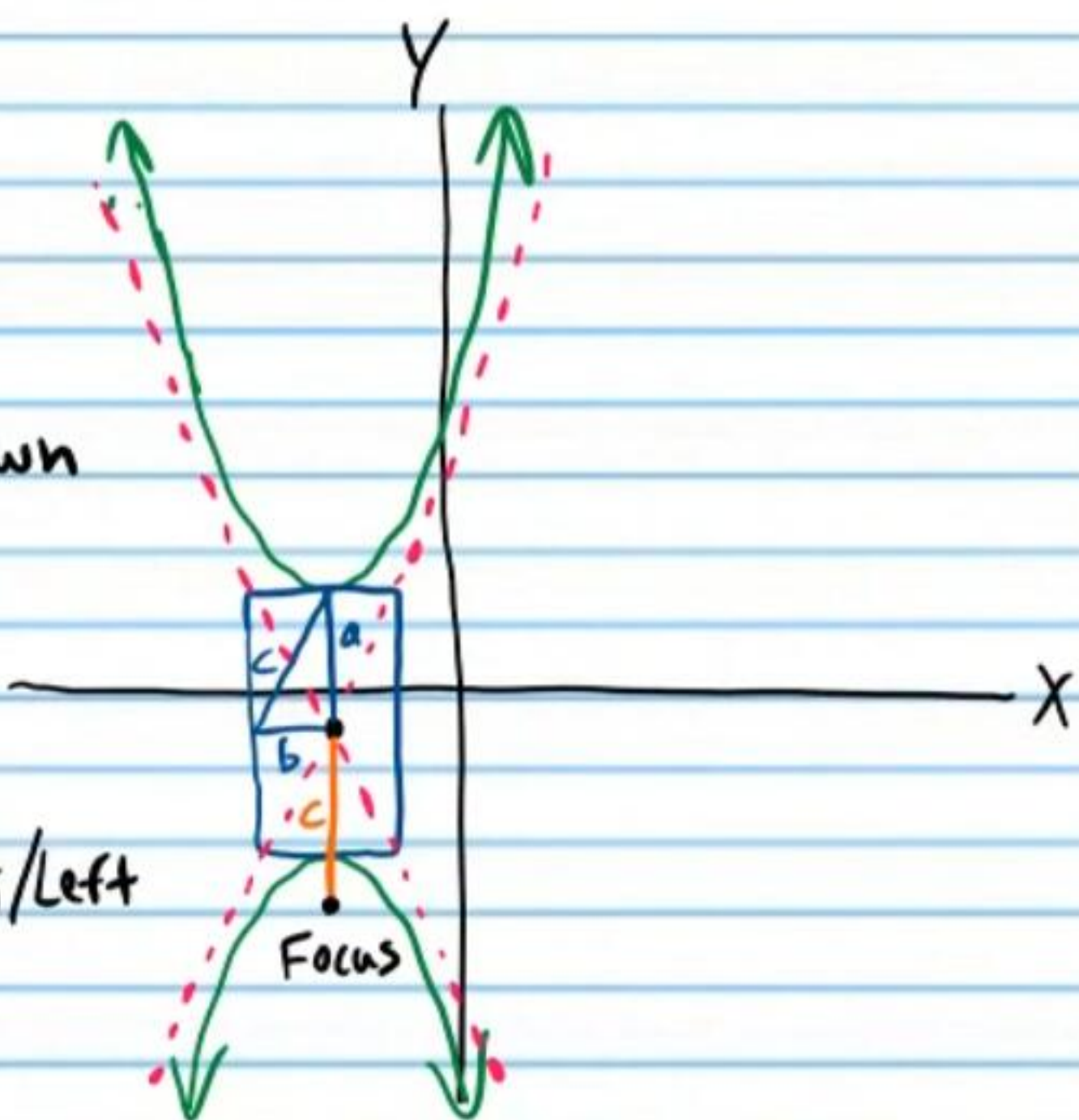
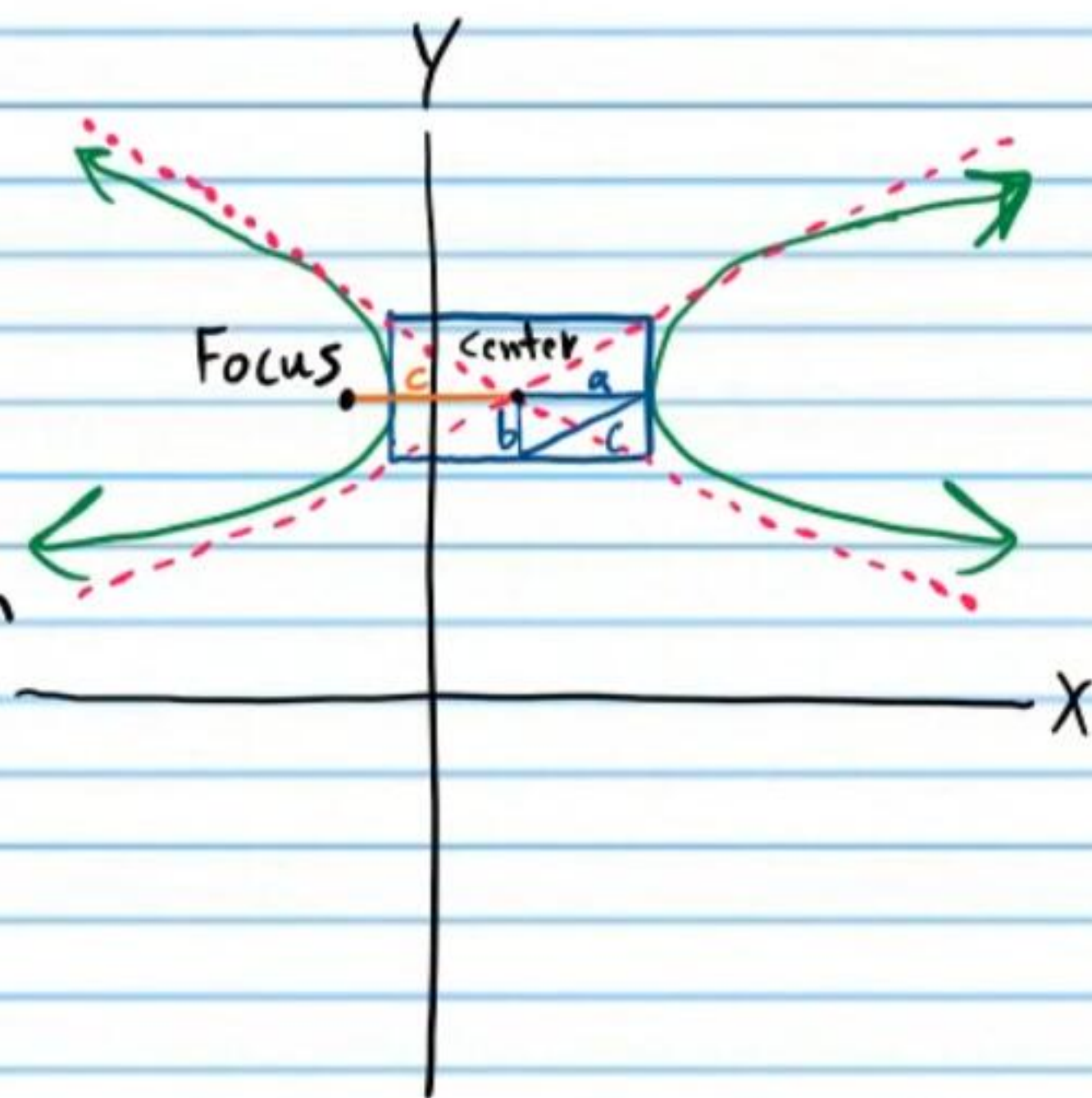
$$y-k = \frac{a}{b}(x-h)$$

$$y-k = -\frac{a}{b}(x-h)$$

Asymptotes Opening Right/Left

$$x-h = \frac{b}{a}(y-k)$$

$$x-h = -\frac{b}{a}(y-k)$$



Writing Equations of Conics Given Their Graphs

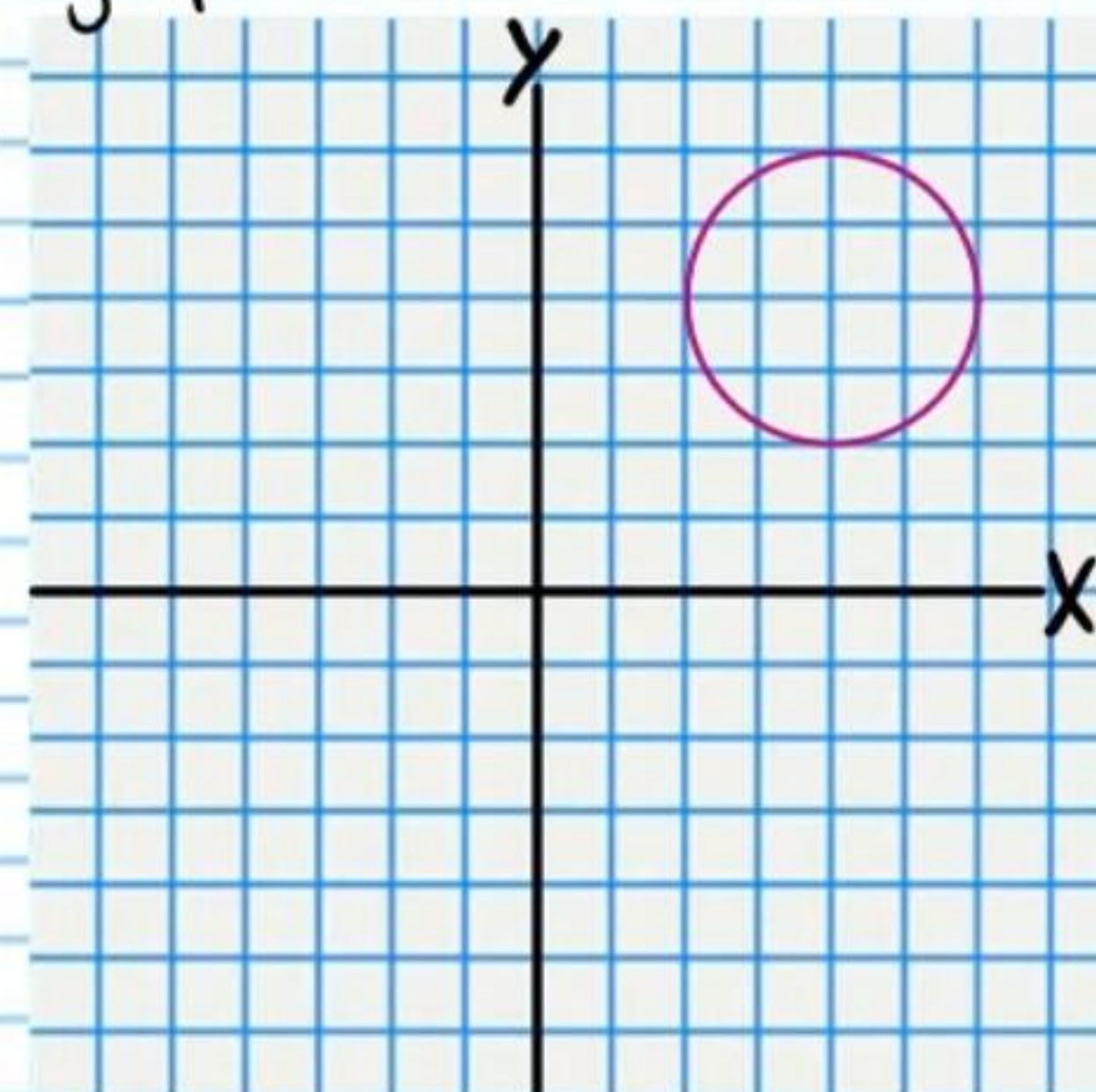
Write an equation of each graph.

- ① $C(h,k) = C(4,4)$, $r=2$

$$(x-h)^2 + (y-k)^2 = r^2$$

$$(x-4)^2 + (y-4)^2 = 2^2$$

$$\boxed{(x-4)^2 + (y-4)^2 = 4}$$

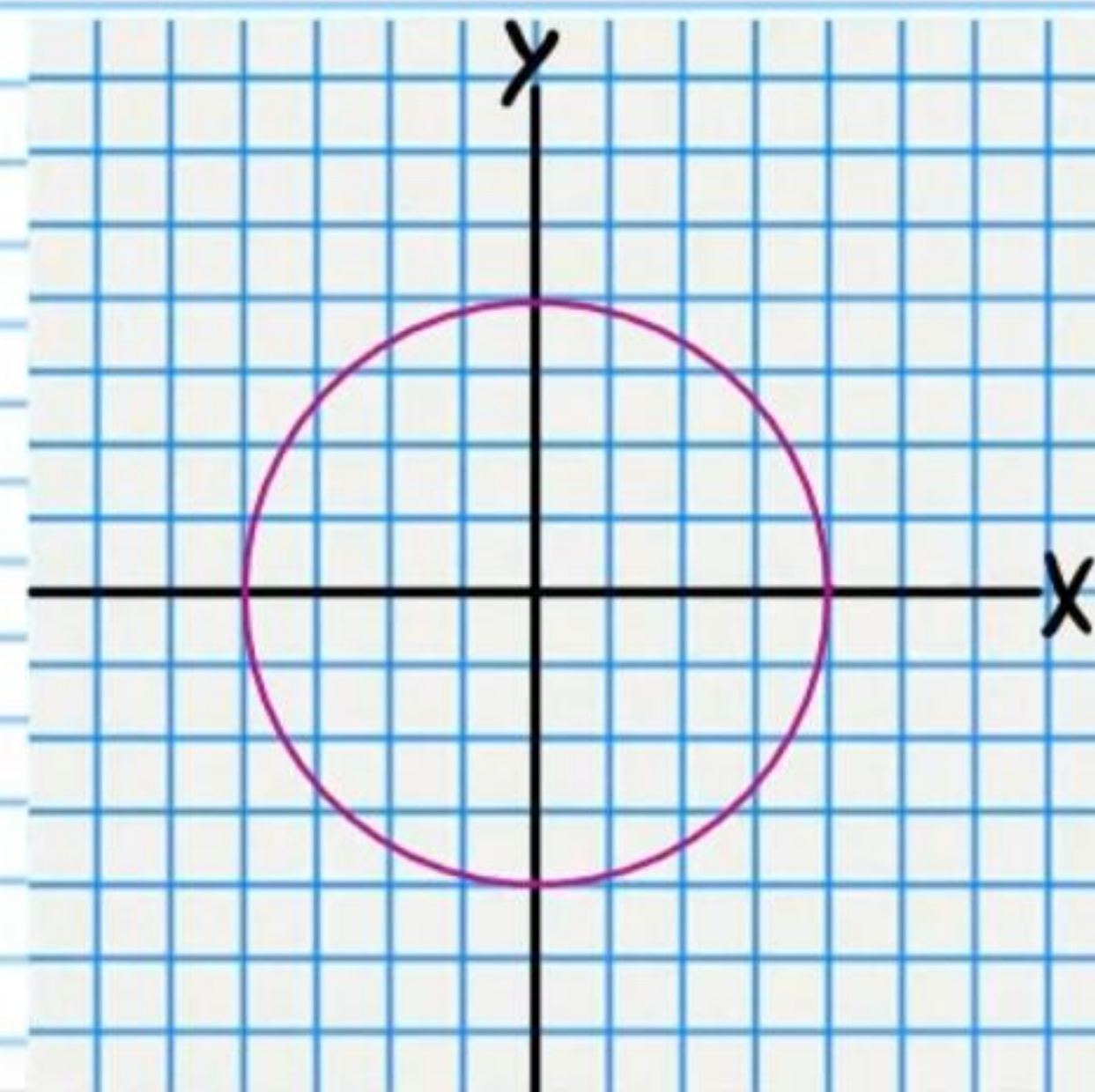


- ② $C(h,k) = C(0,0)$, $r=4$

$$(x-h)^2 + (y-k)^2 = r^2$$

$$(x-0)^2 + (y-0)^2 = 4^2$$

$$\boxed{x^2 + y^2 = 16}$$

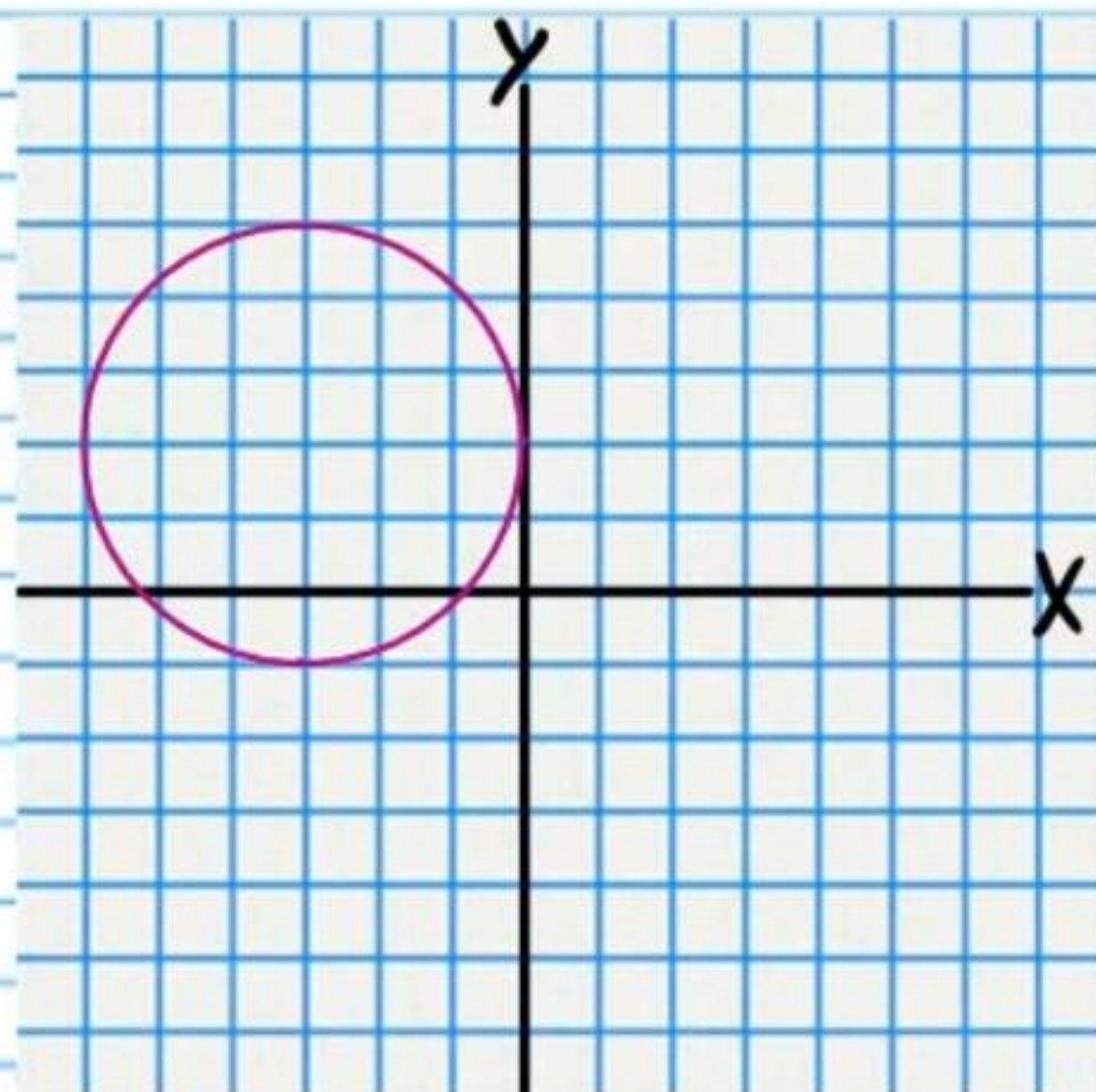


• ③ $C(h,k) = C(-3,2), r=3$

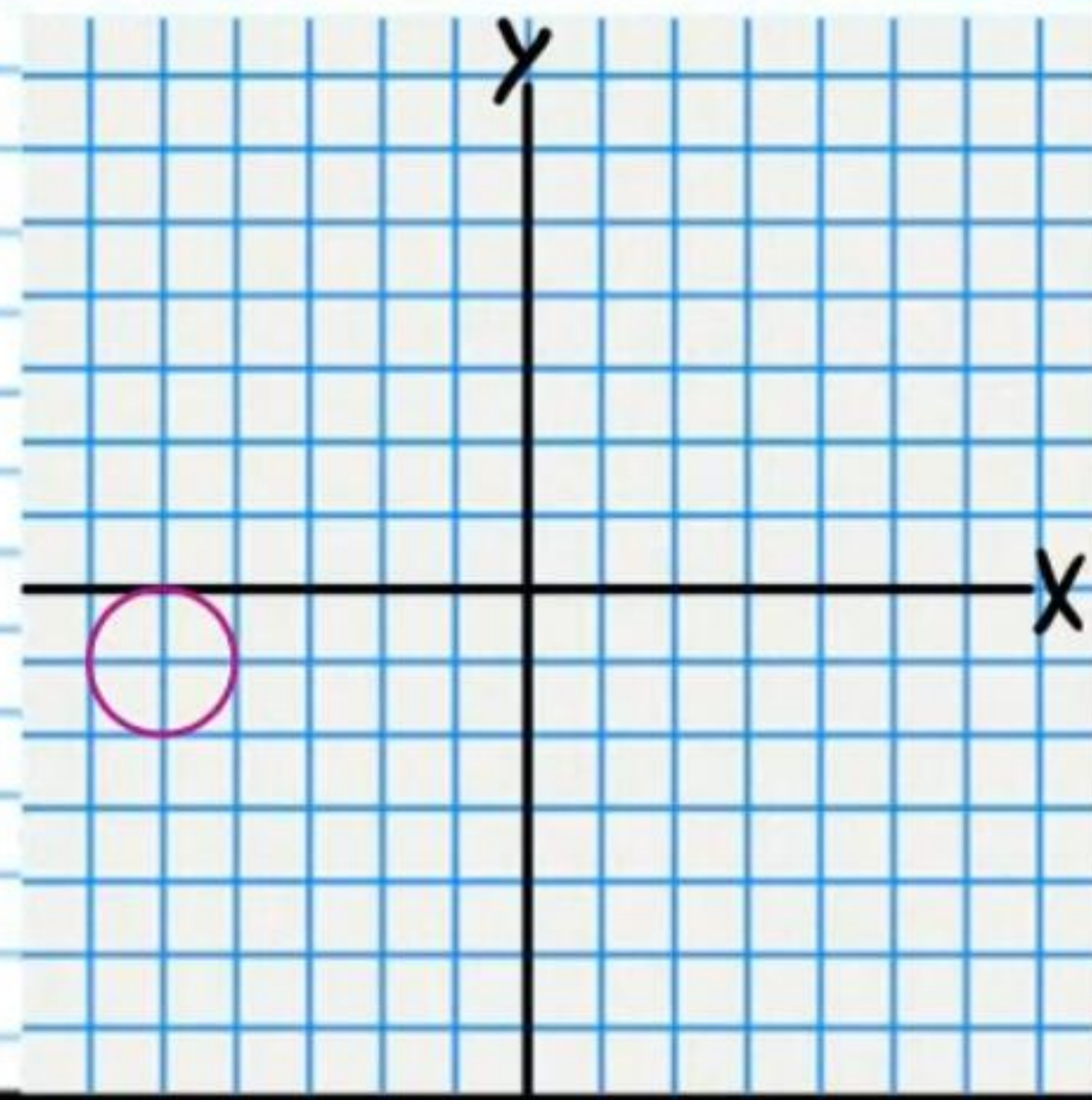
$$(x-h)^2 + (y-k)^2 = r^2$$

$$(x-(-3))^2 + (y-2)^2 = 3^2$$

$$\boxed{(x+3)^2 + (y-2)^2 = 9}$$



④ $\boxed{(x+5)^2 + (y+1)^2 = 1}$



"P" is a point on each parabola and "V" is the vertex.

• ⑤ $P(0,1) \quad V(2,4)$

$$y-k = a(x-h)^2$$

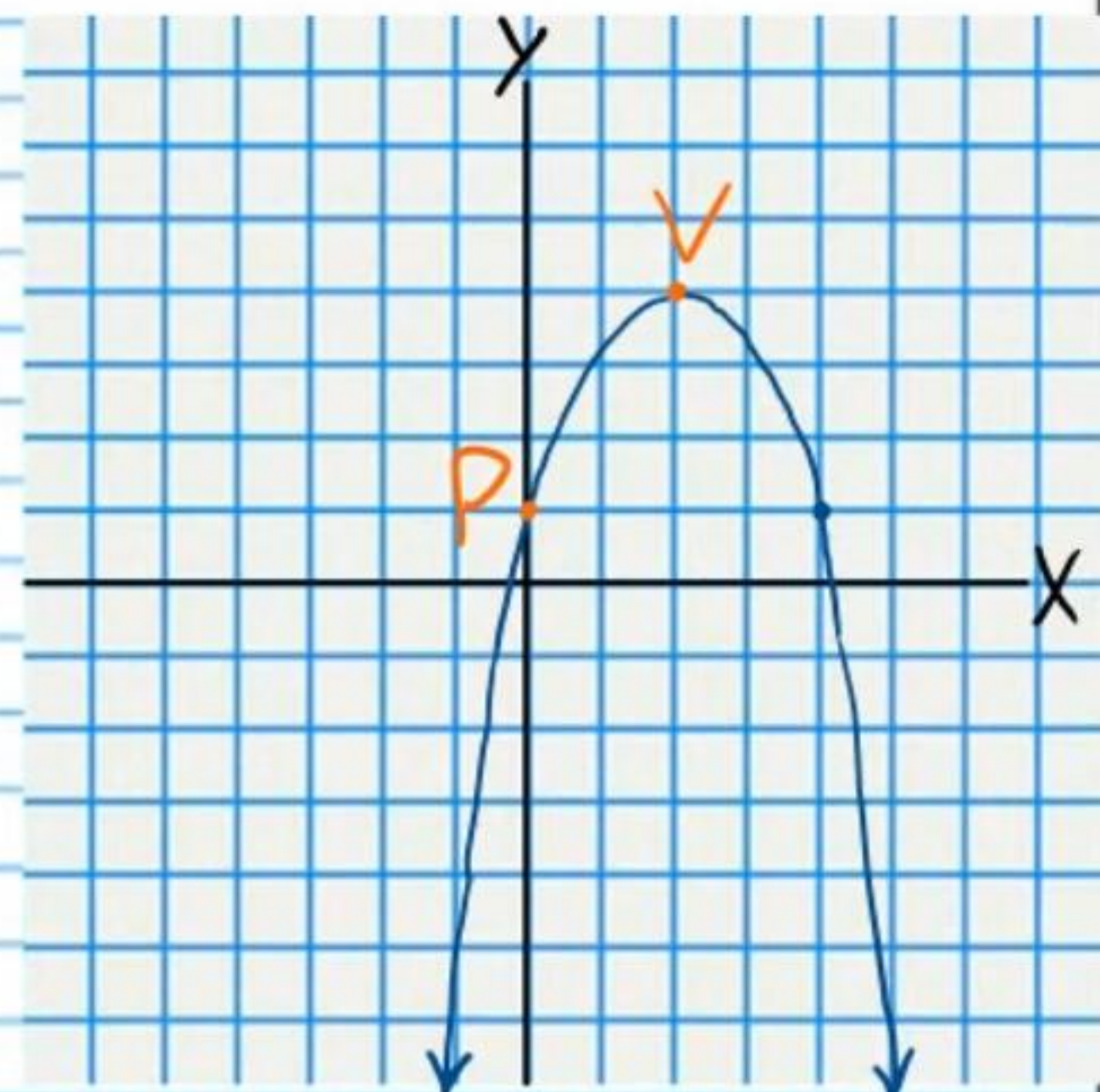
$$1-4 = a(0-2)^2$$

$$-3 = a(-2)^2$$

$$-3 = 4a$$

$$-\frac{3}{4} = a$$

$$\boxed{y-4 = -\frac{3}{4}(x-2)^2}$$



⑥ $P(2,-3) \quad V(1,-4)$

$$x-h = a(y-k)^2$$

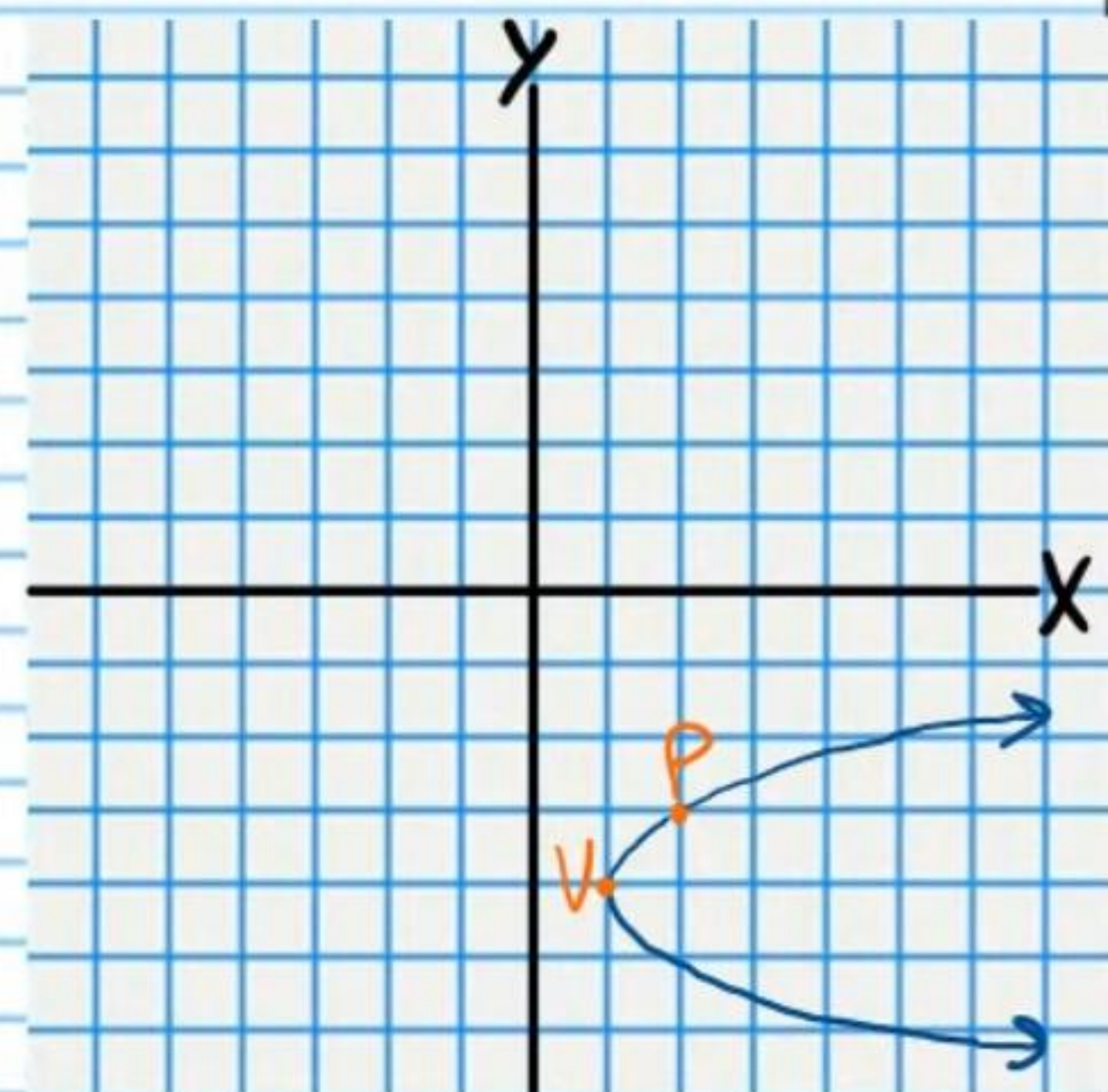
$$2-1 = a(-3-(-4))^2$$

$$1 = a(-3+4)^2$$

$$1 = a(1)^2$$

$$a = 1$$

$$\boxed{x-1 = (y+4)^2}$$



• ⑦ $P(-2, 5)$ $V(-3, 3)$

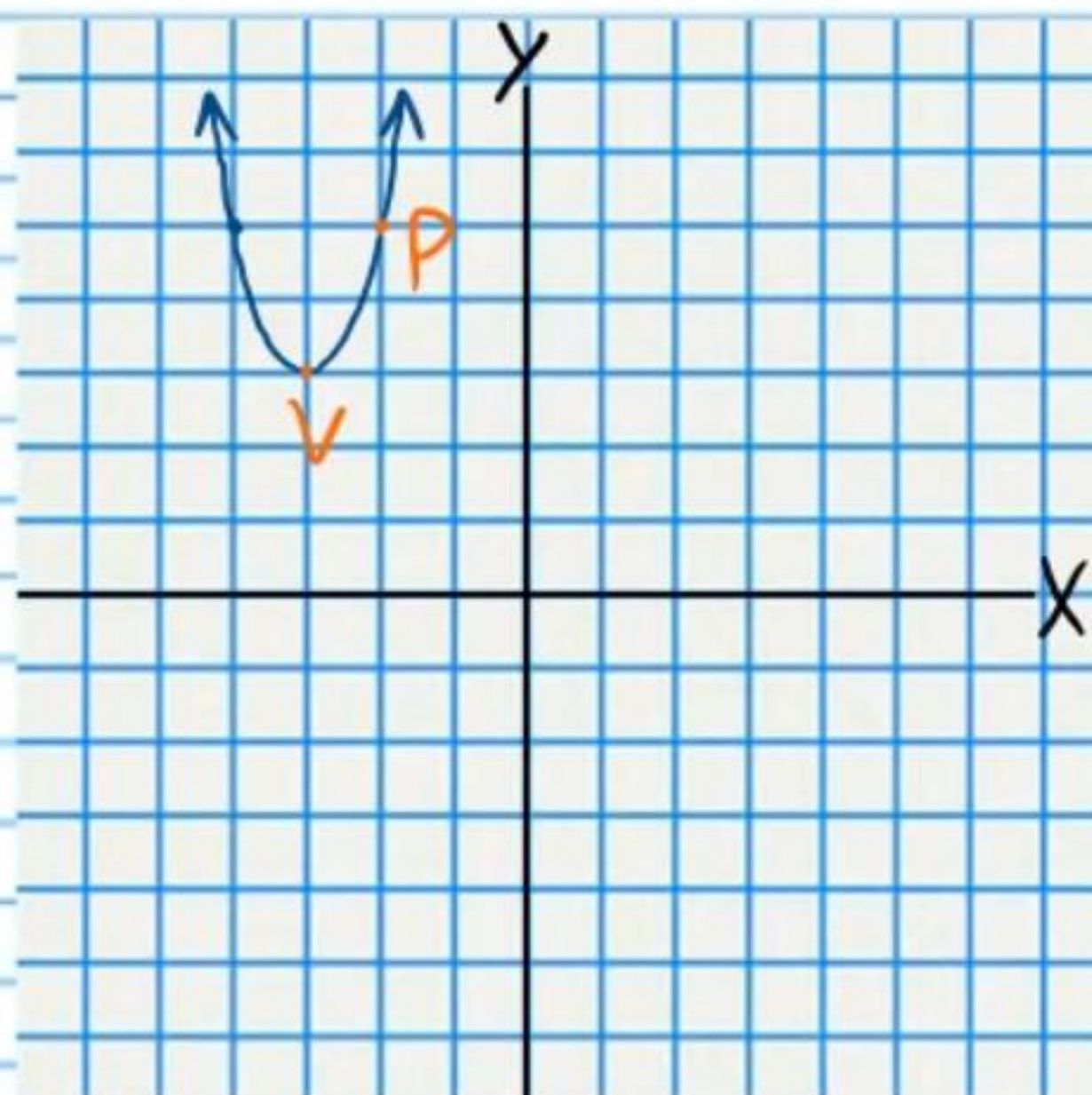
$$y - k = a(x - h)^2$$

$$5 - 3 = a(-2 - (-3))^2$$

$$2 = a(-2 + 3)^2$$

$$a = 2$$

$$y - 3 = 2(x + 3)^2$$



• ⑧ $P(-3, 2)$ $V(-2, -2)$

$$x - h = a(y - k)^2$$

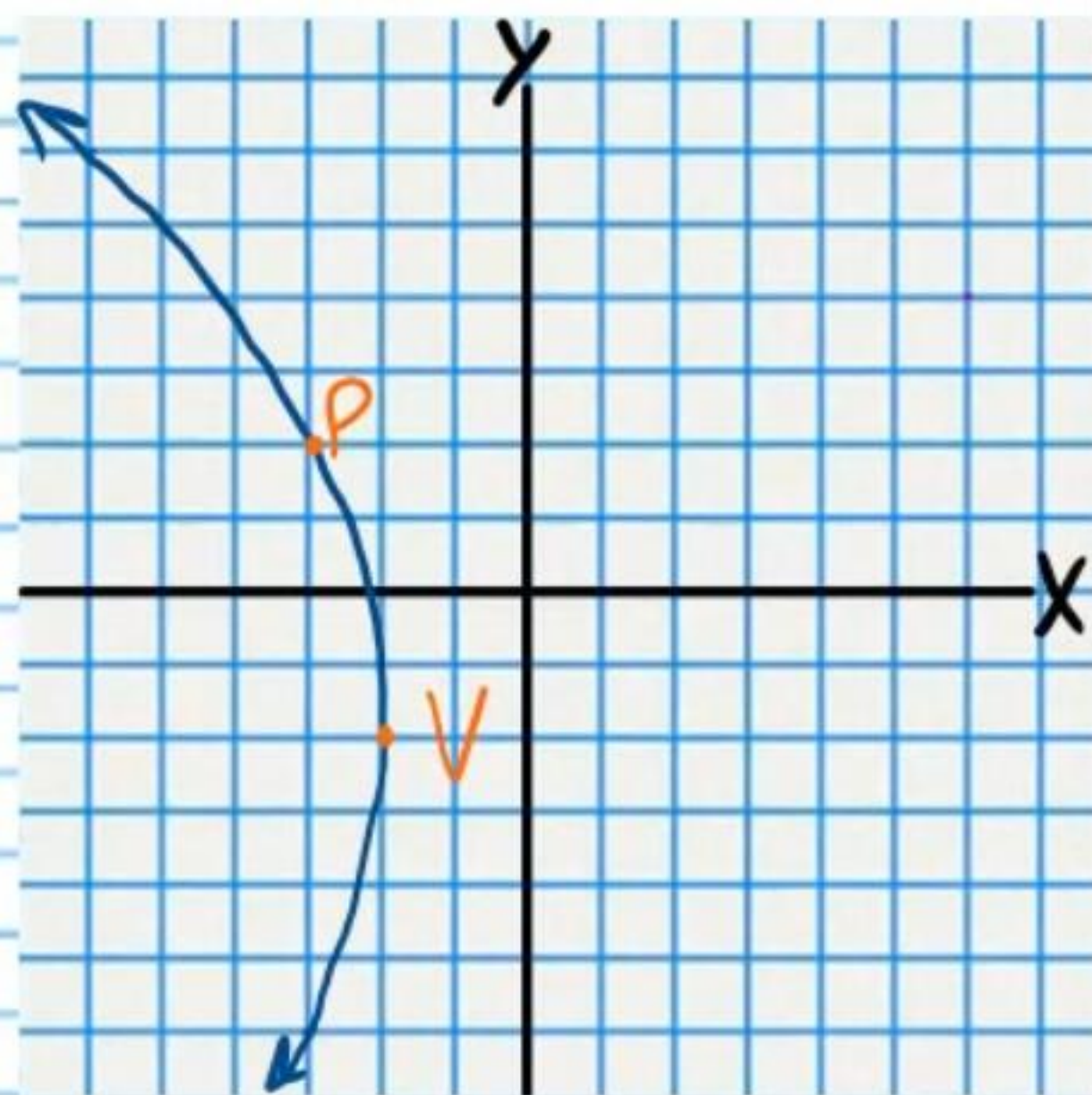
$$-3 - (-2) = a(2 - (-2))^2$$

$$-1 = a(4)^2$$

$$a = -\frac{1}{16}$$

$$x - (-2) = -\frac{1}{16}(y - (-2))^2$$

$$x + 2 = -\frac{1}{16}(y + 2)^2$$

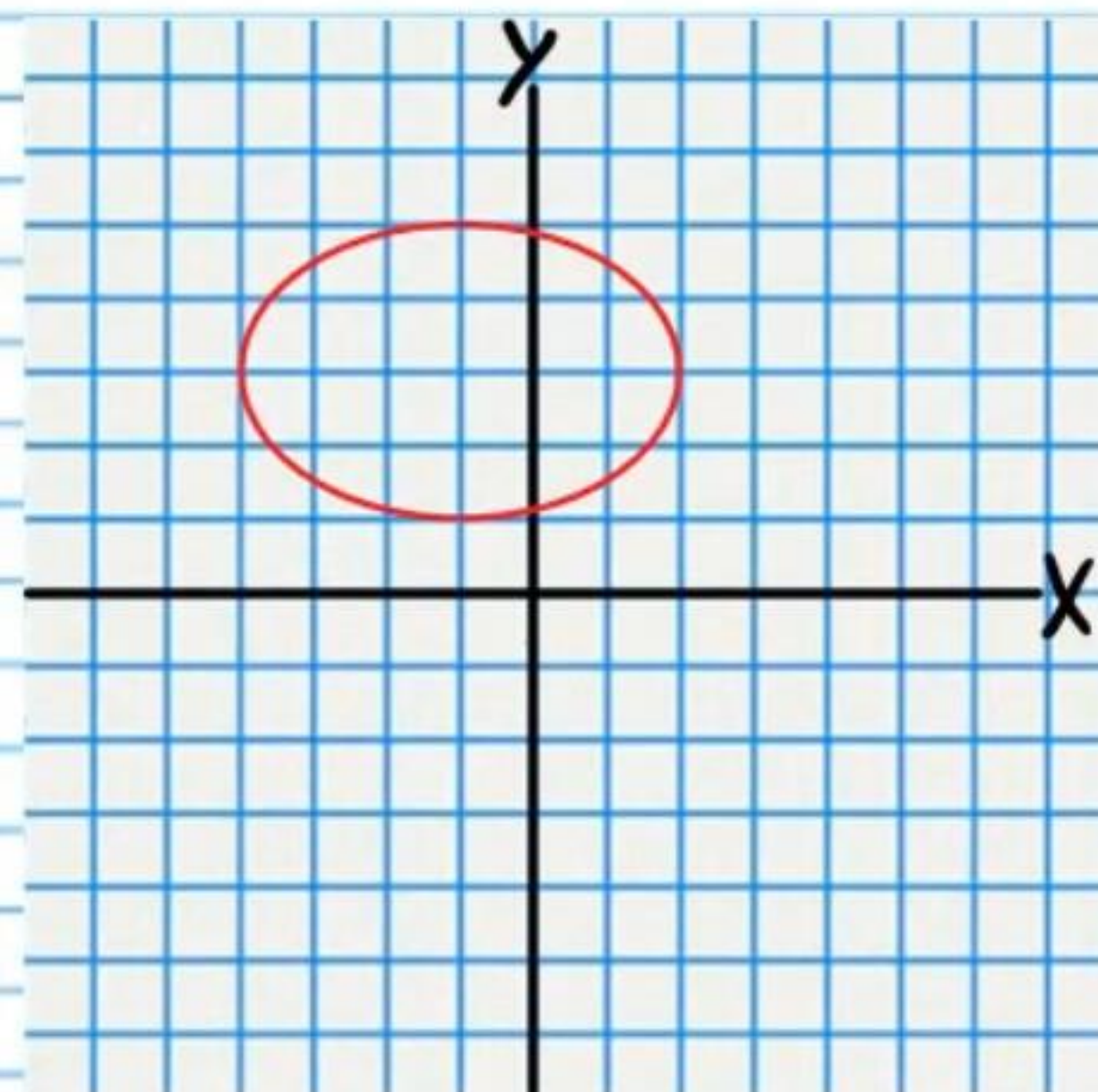


• ⑨ $C(-1, 3)$

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1$$

$$\frac{(x - (-1))^2}{3^2} + \frac{(y - 3)^2}{2^2} = 1$$

$$\frac{(x + 1)^2}{9} + \frac{(y - 3)^2}{4} = 1$$

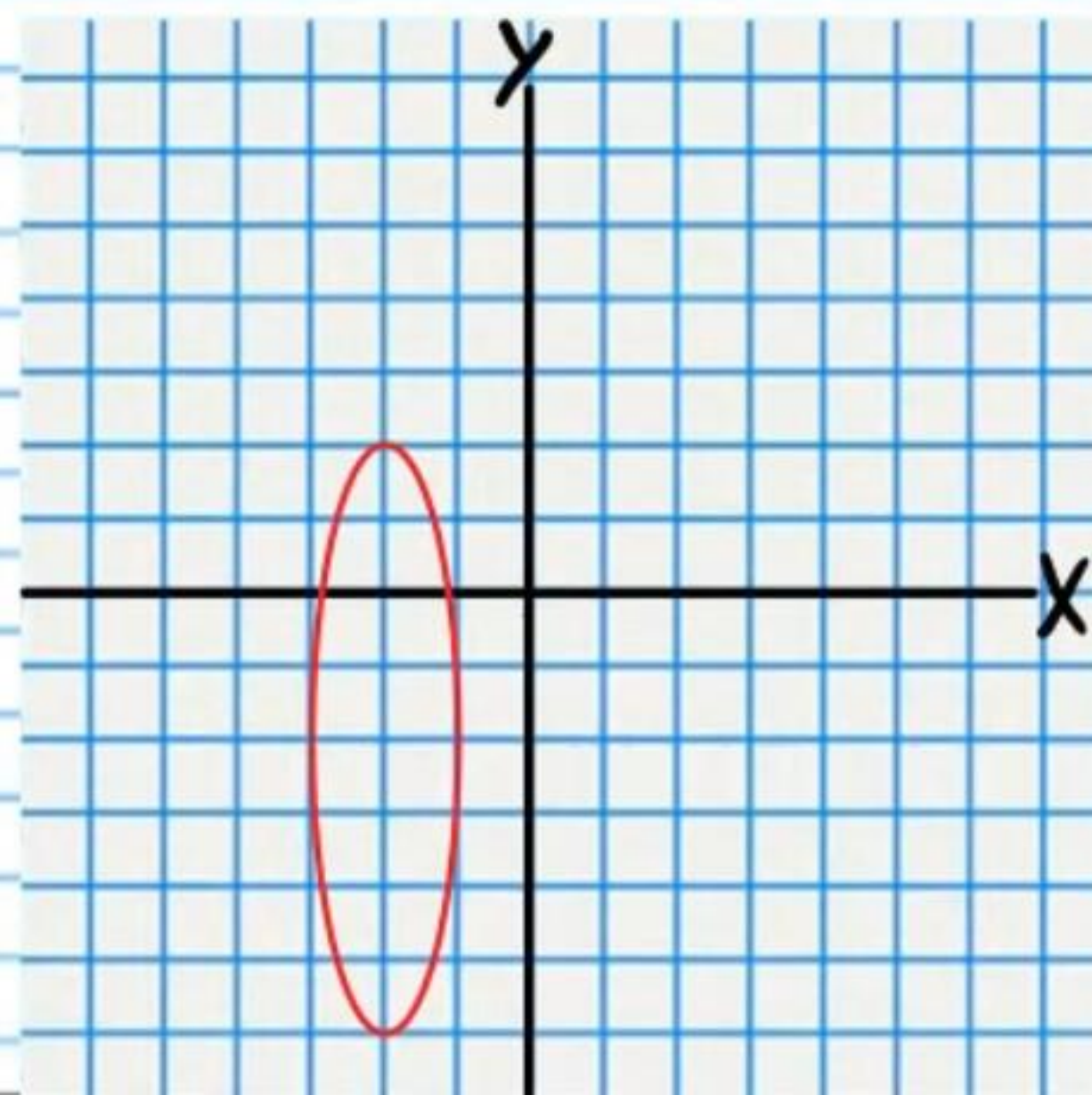


• ⑩ $C(-2, -2)$

$$\frac{(y - k)^2}{a^2} + \frac{(x - h)^2}{b^2} = 1$$

$$\frac{(y - (-2))^2}{4^2} + \frac{(x - (-2))^2}{1^2} = 1$$

$$\frac{(y + 2)^2}{16} + (x + 2)^2 = 1$$

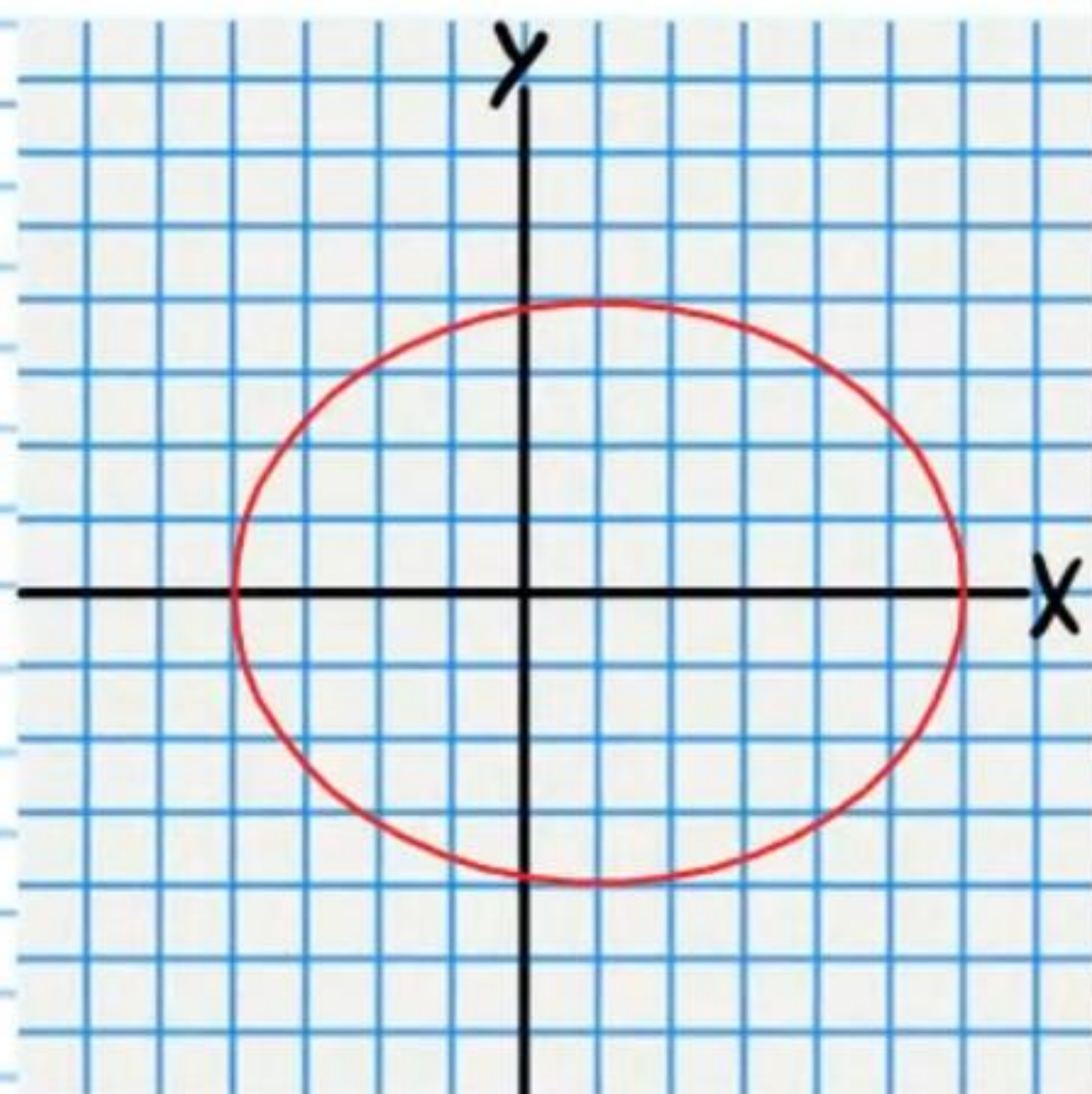


• ⑪ $C(1,0)$

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

$$\frac{(x-1)^2}{5^2} + \frac{(y-0)^2}{4^2} = 1$$

$$\boxed{\frac{(x-1)^2}{25} + \frac{y^2}{16} = 1}$$

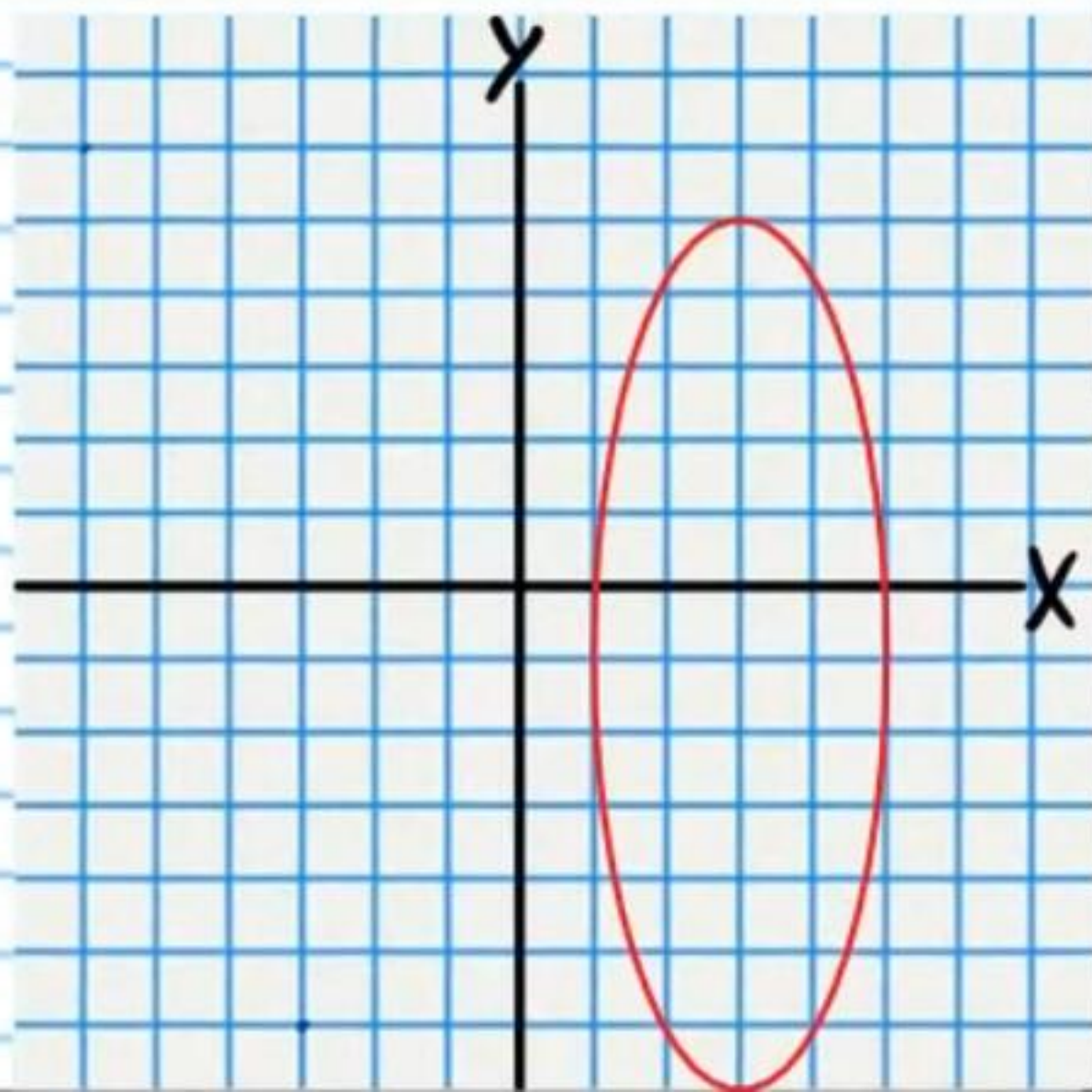


• ⑫ $C(3,-1)$

$$\frac{(y-k)^2}{a^2} + \frac{(x-h)^2}{b^2} = 1$$

$$\frac{(y-(-1))^2}{6^2} + \frac{(x-3)^2}{2^2} = 1$$

$$\boxed{\frac{(y+1)^2}{36} + \frac{(x-3)^2}{4} = 1}$$

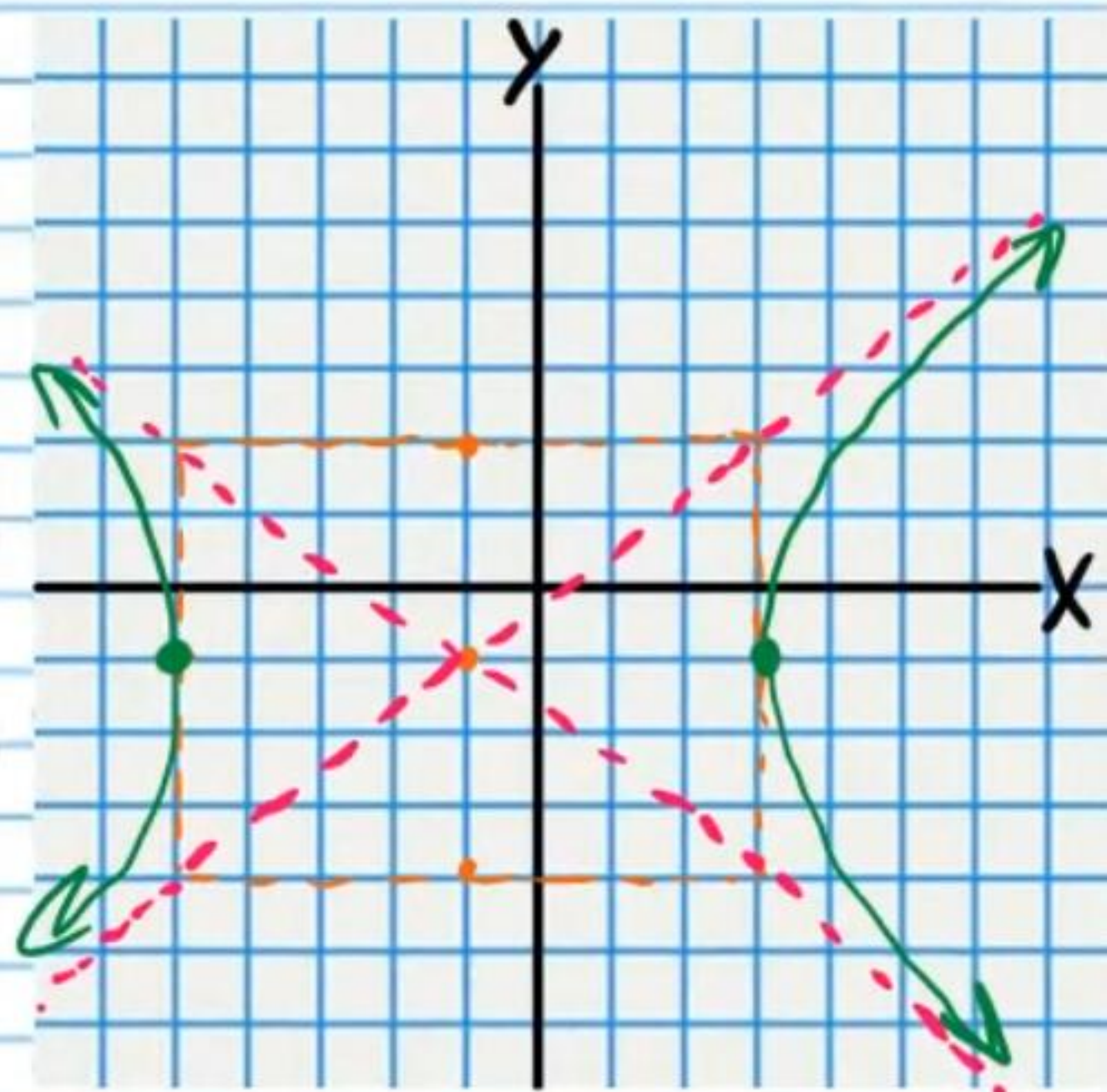


• ⑬ $C(-1,-1)$

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

$$\frac{(x-(-1))^2}{4^2} - \frac{(y-(-1))^2}{3^2} = 1$$

$$\boxed{\frac{(x+1)^2}{16} - \frac{(y+1)^2}{9} = 1}$$

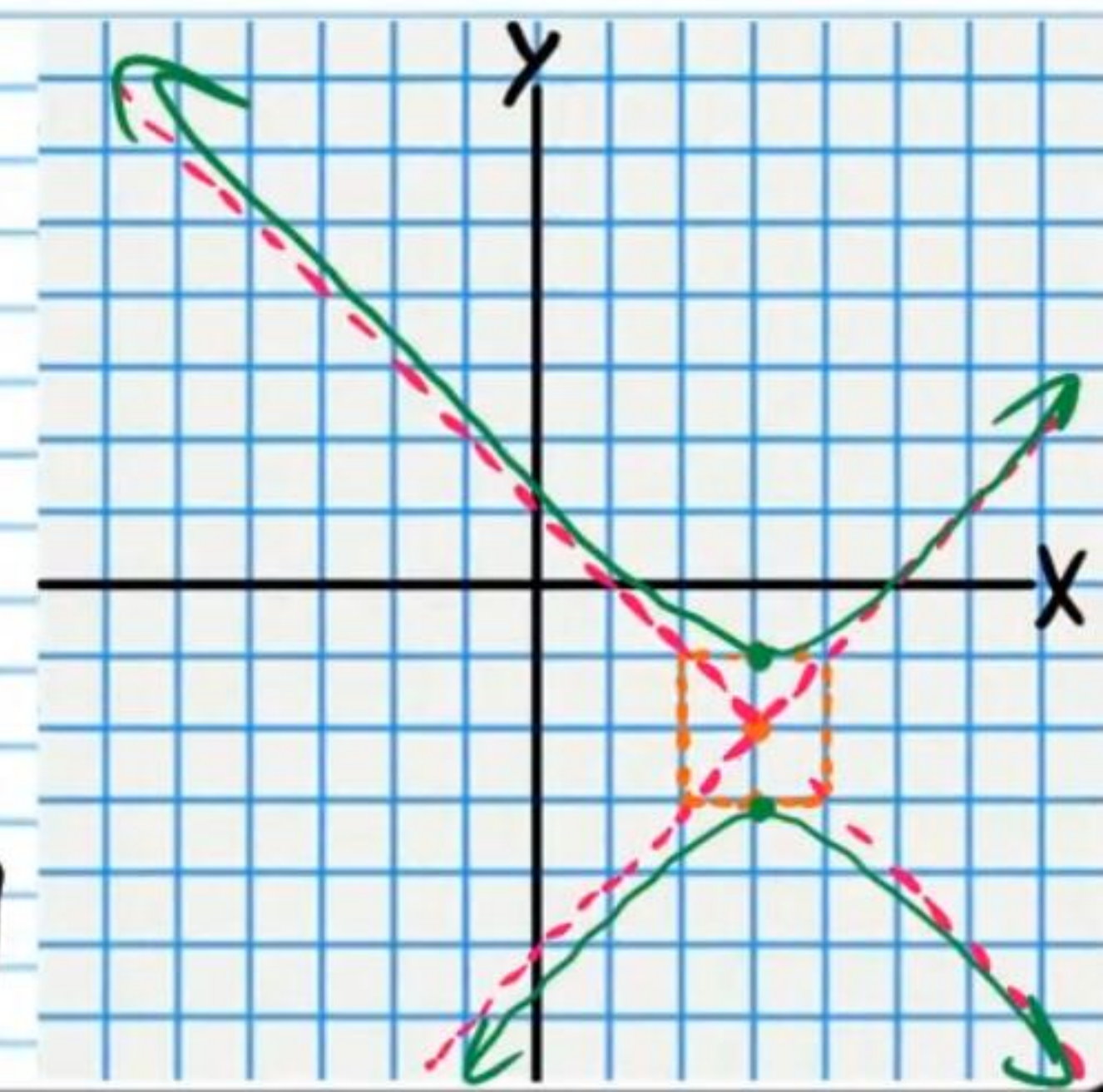


• ⑭ $C(3,-2)$

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

$$\frac{(y-(-2))^2}{1^2} - \frac{(x-3)^2}{1^2} = 1$$

$$\boxed{(y+2)^2 - (x-3)^2 = 1}$$

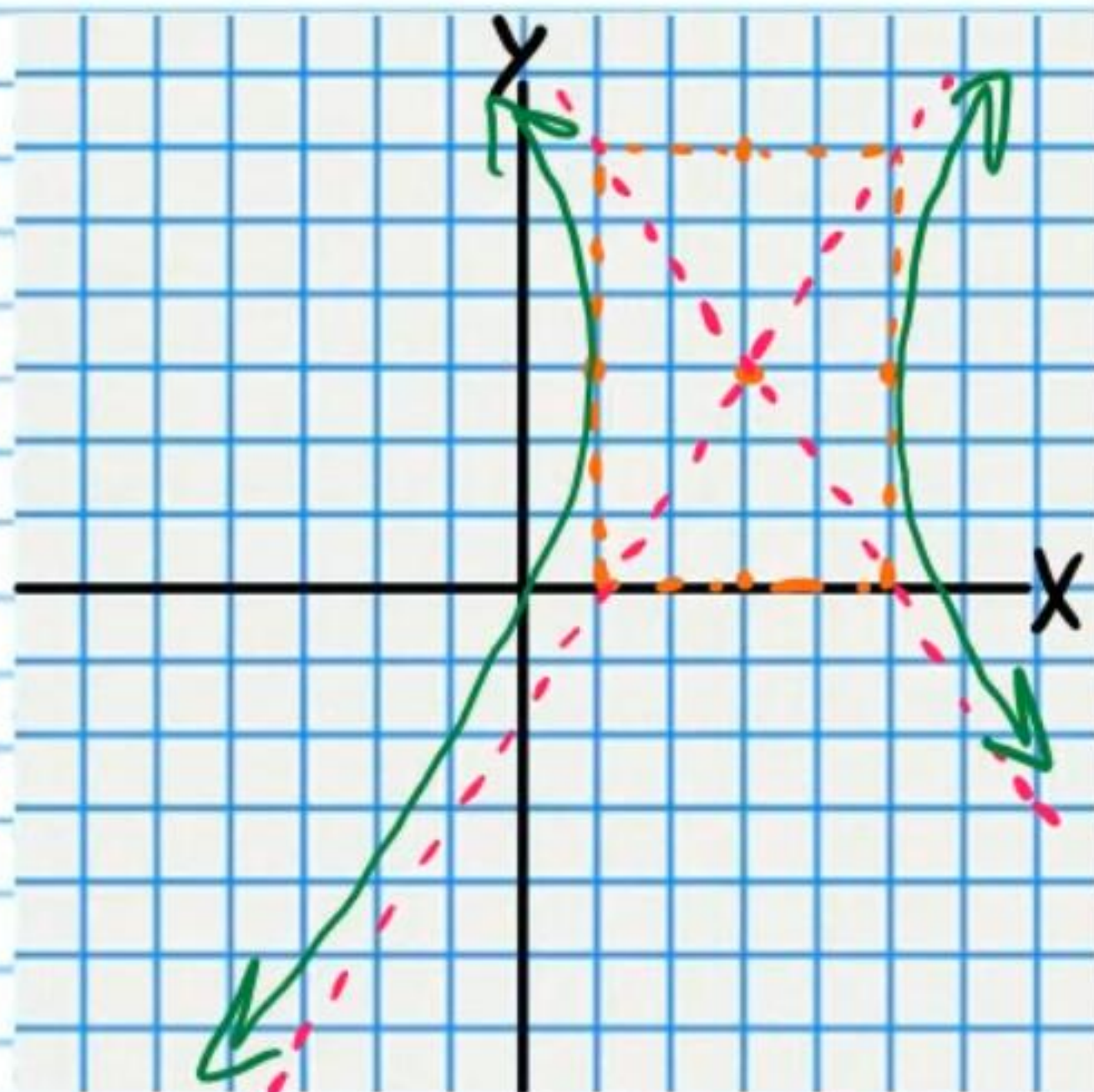


• ⑮ $C(3,3)$

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

$$\frac{(x-3)^2}{2^2} - \frac{(y-3)^2}{3^2} = 1$$

$$\boxed{\frac{(x-3)^2}{4} - \frac{(y-3)^2}{9} = 1}$$

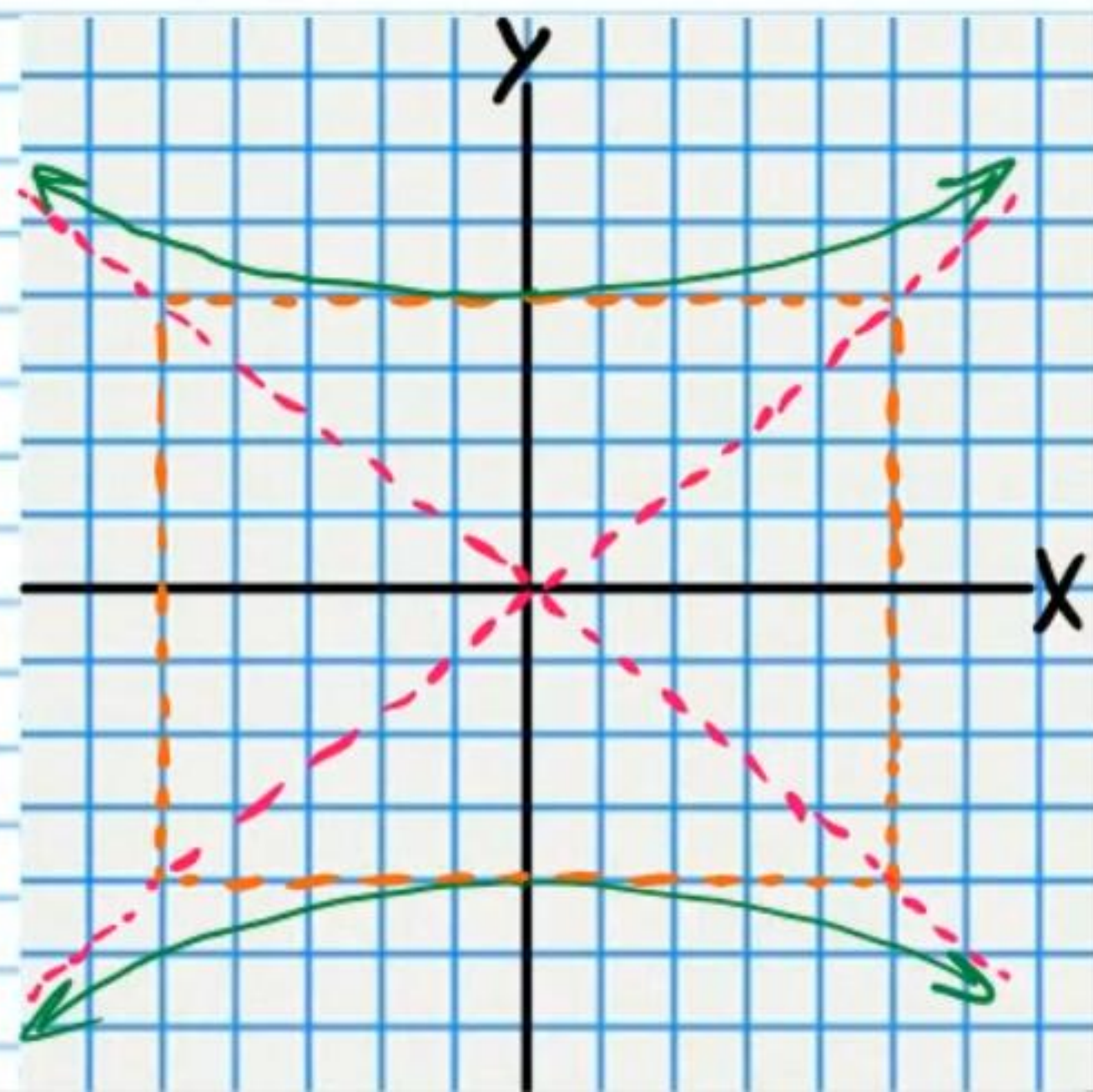


• ⑯ $C(0,0)$

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

$$\frac{(y-0)^2}{4^2} - \frac{(x-0)^2}{5^2} = 1$$

$$\boxed{\frac{y^2}{16} - \frac{x^2}{25} = 1}$$



Writing and Graphing Equations of Conics Given Various Information

Find equations of circles with the following characteristics and graph each equation.

• ⑰ Center at $(3, -1)$ and point on the circle $(0, -2)$.

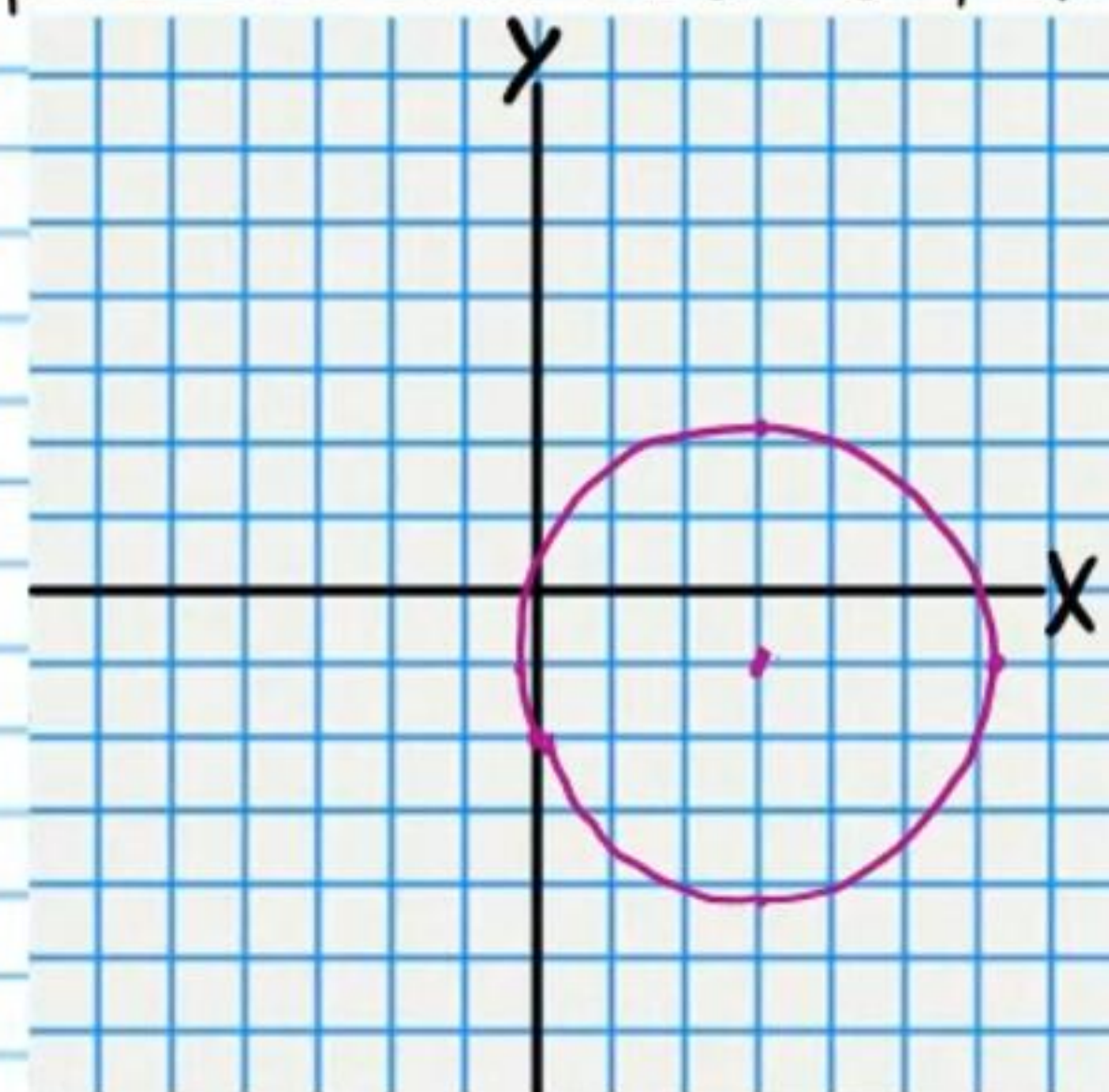
$$(x-h)^2 + (y-k)^2 = r^2$$

$$(0-3)^2 + (-2-(-1))^2 = r^2$$

$$(-3)^2 + (-1)^2 = r^2$$

$$10 = r^2$$

$$\boxed{(x-3)^2 + (y+1)^2 = 10}$$



⑱ Center on the line $y=3$, radius 1, and point on the circle $(-5, 2)$.

$$(x-h)^2 + (y-k)^2 = r^2$$

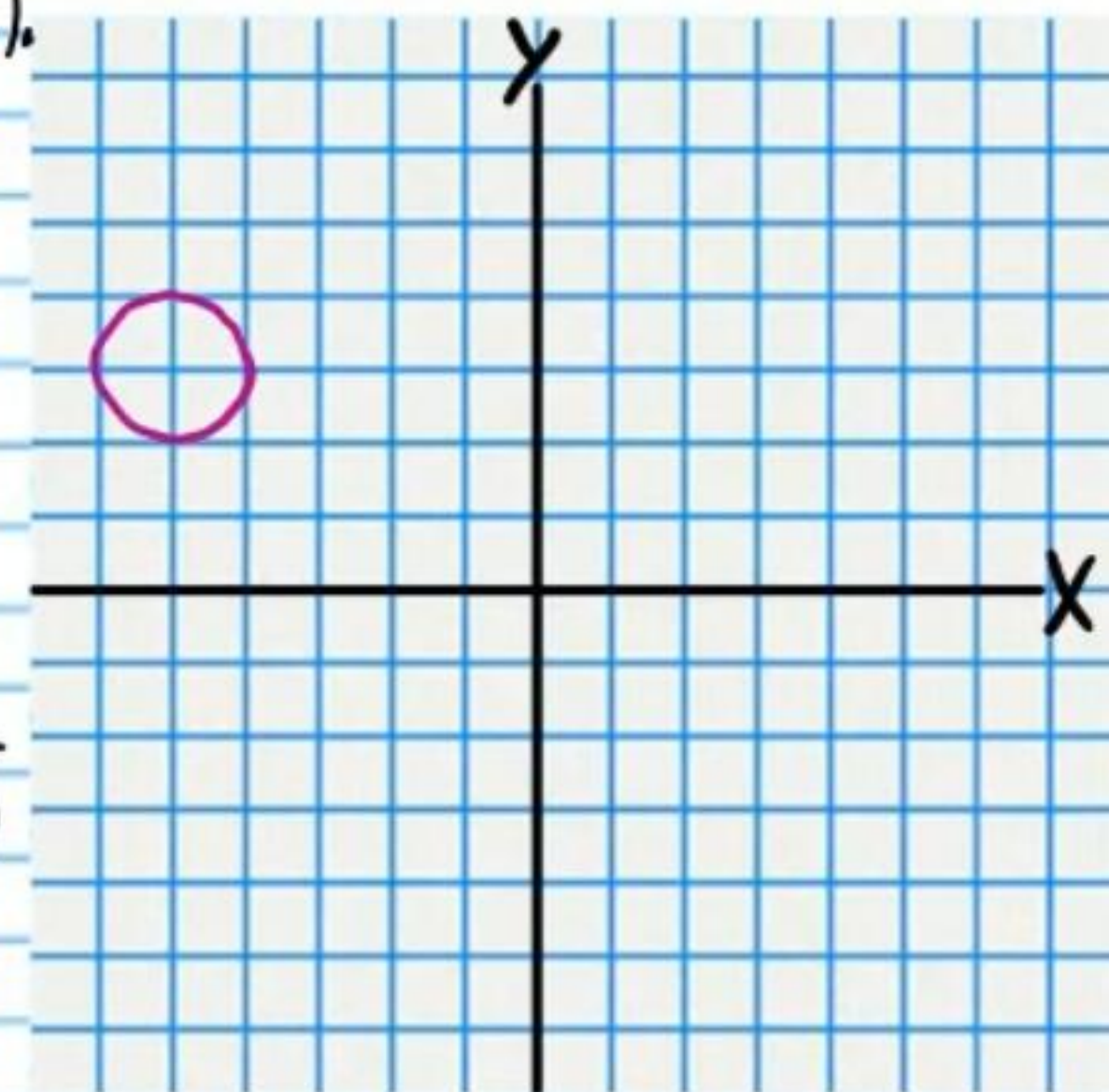
$$(-5-h)^2 + (2-3)^2 = 1^2$$

$$(-5-h)^2 + (-1)^2 = 1$$

$$(-5-h)^2 = 0$$

$$-5-h = \pm\sqrt{0}$$

$$\boxed{(x+5)^2 + (y-3)^2 = 1} \quad h = -5$$



- (19) Center at $(2, 4)$ and point on the circle $(1, 2)$.

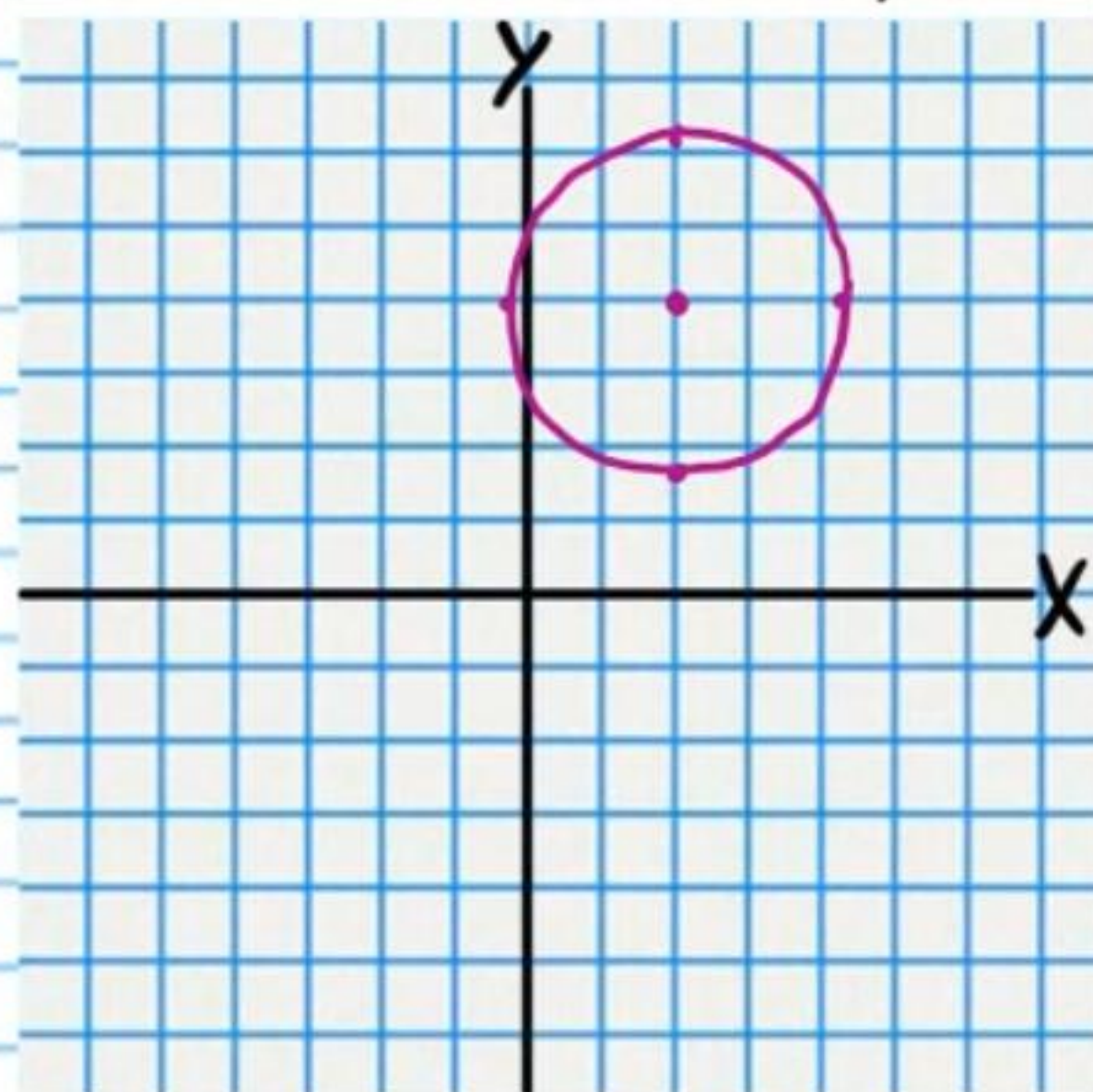
$$(x-h)^2 + (y-k)^2 = r^2$$

$$(1-2)^2 + (2-4)^2 = r^2$$

$$(-1)^2 + (-2)^2 = r^2$$

$$5 = r^2$$

$$\boxed{(x-2)^2 + (y-4)^2 = 5}$$



- (20) Center on the line $x = -3$, radius 4, and point on the circle $(1, 0)$.

$$(x-h)^2 + (y-k)^2 = r^2$$

$$(1-(-3))^2 + (0-k)^2 = 4^2$$

$$4^2 + (-k)^2 = 16$$

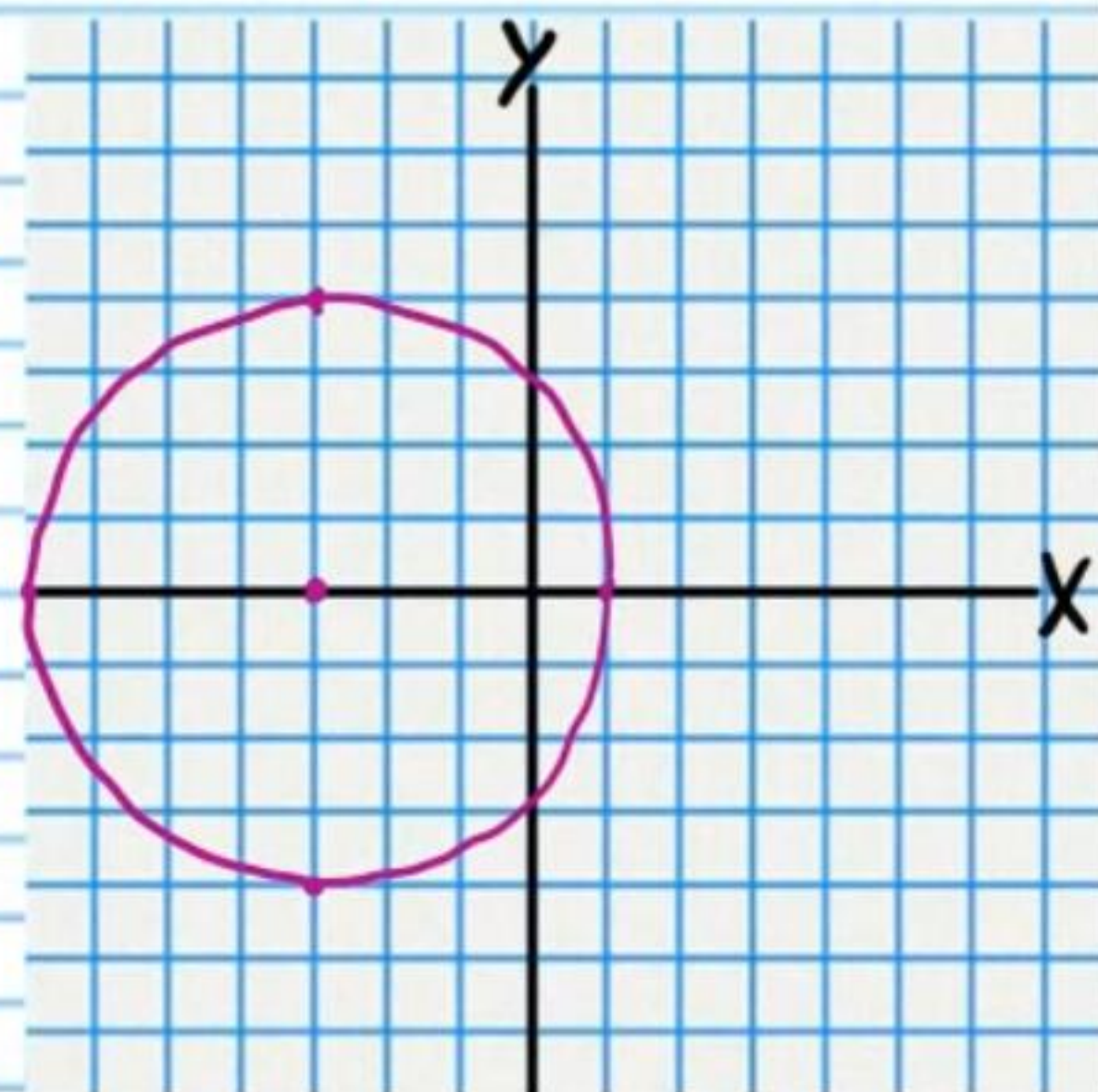
$$k^2 = 0$$

$$k = \pm\sqrt{0}$$

$$k = 0$$

$$(x+3)^2 + (y-0)^2 = 4^2$$

$$\boxed{(x+3)^2 + y^2 = 16}$$



Find equations of parabolas with the following characteristics and graph each equation.

- (21) Vertex $(-4, 5)$ and focus $(-3, 5)$.

$$p = \frac{1}{4a}$$

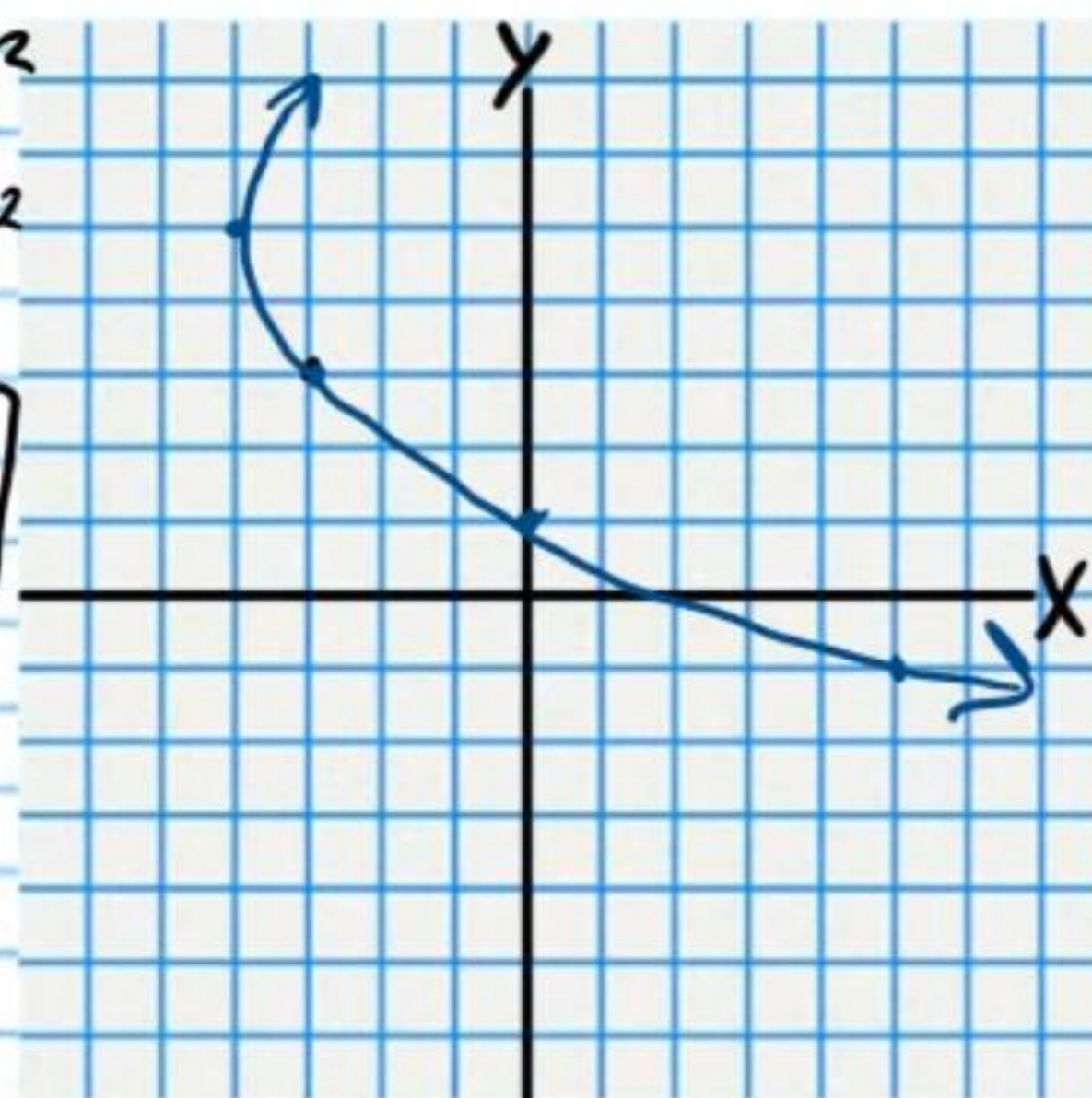
$$1 = \frac{1}{4a}$$

$$a = \frac{1}{4}$$

$$x-h = a(y-k)^2$$

$$x-(-4) = \frac{1}{4}(y-5)^2$$

$$\boxed{x+4 = \frac{1}{4}(y-5)^2}$$



- (22) Vertex $(4, 1)$ and focus $(4, \frac{1}{2})$.

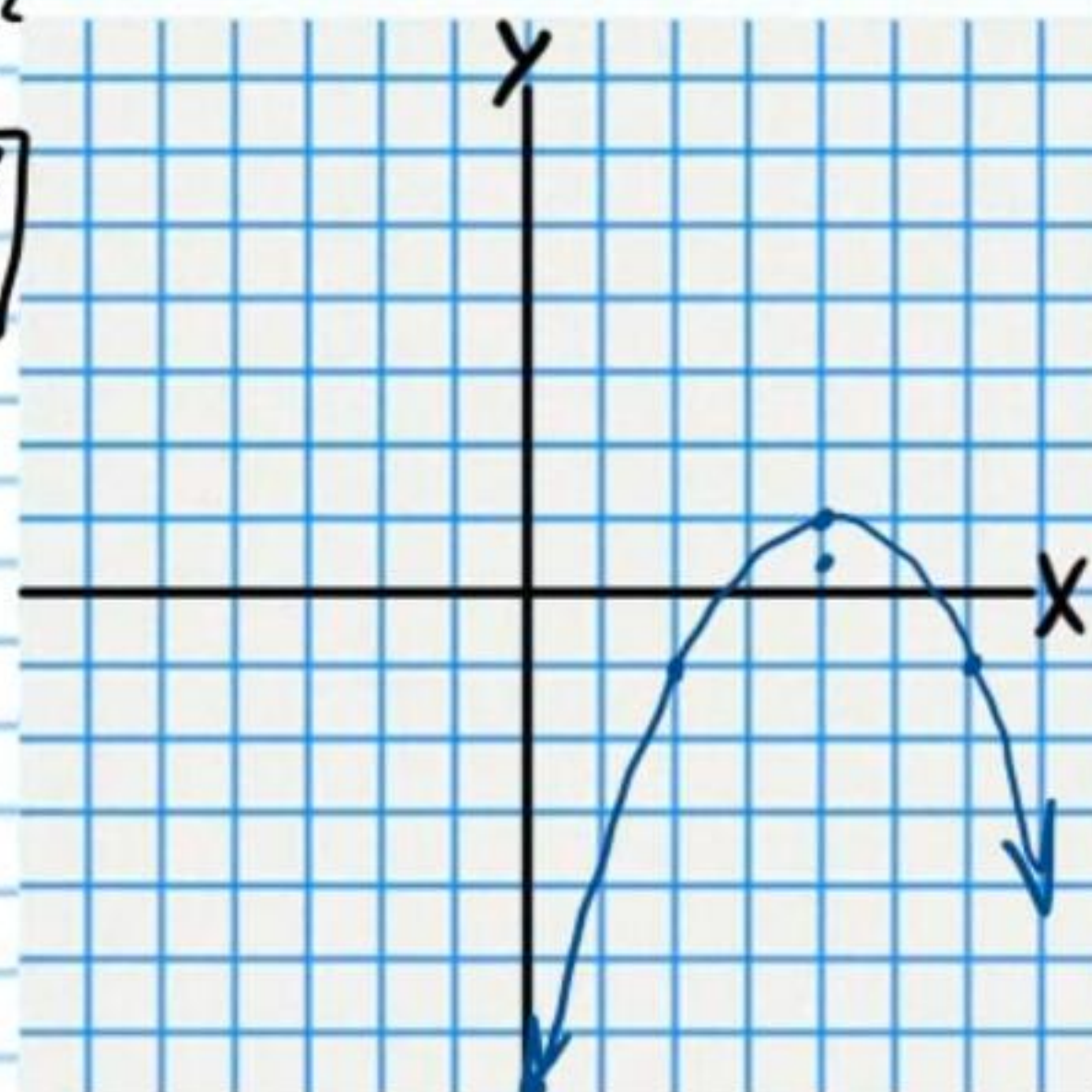
$$p = \frac{1}{4a}$$

$$\frac{1}{2} = \frac{1}{4a}$$

$$a = \frac{1}{2}$$

$$y-k = a(x-h)^2$$

$$\boxed{y-1 = -\frac{1}{2}(x-4)^2}$$



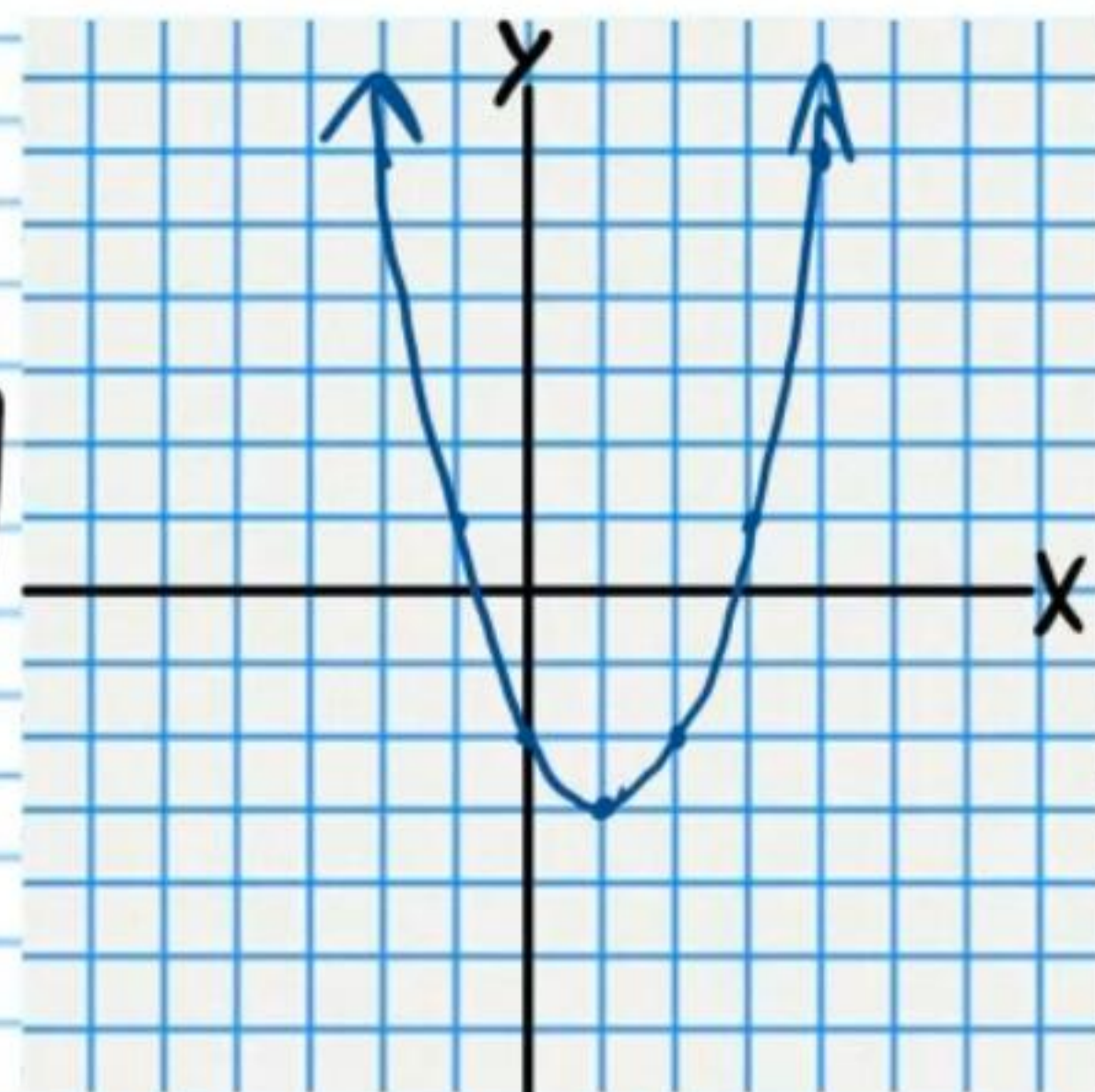
- (23) Vertex $(1, -3)$ and focus $(1, -2\frac{3}{4})$.

$$P = \frac{1}{4a} \quad y - k = a(x - h)^2$$

$$\frac{1}{4} = \frac{1}{4a} \quad y + 3 = 1(x - 1)^2$$

$$a = 1$$

$$\boxed{y + 3 = (x - 1)^2}$$



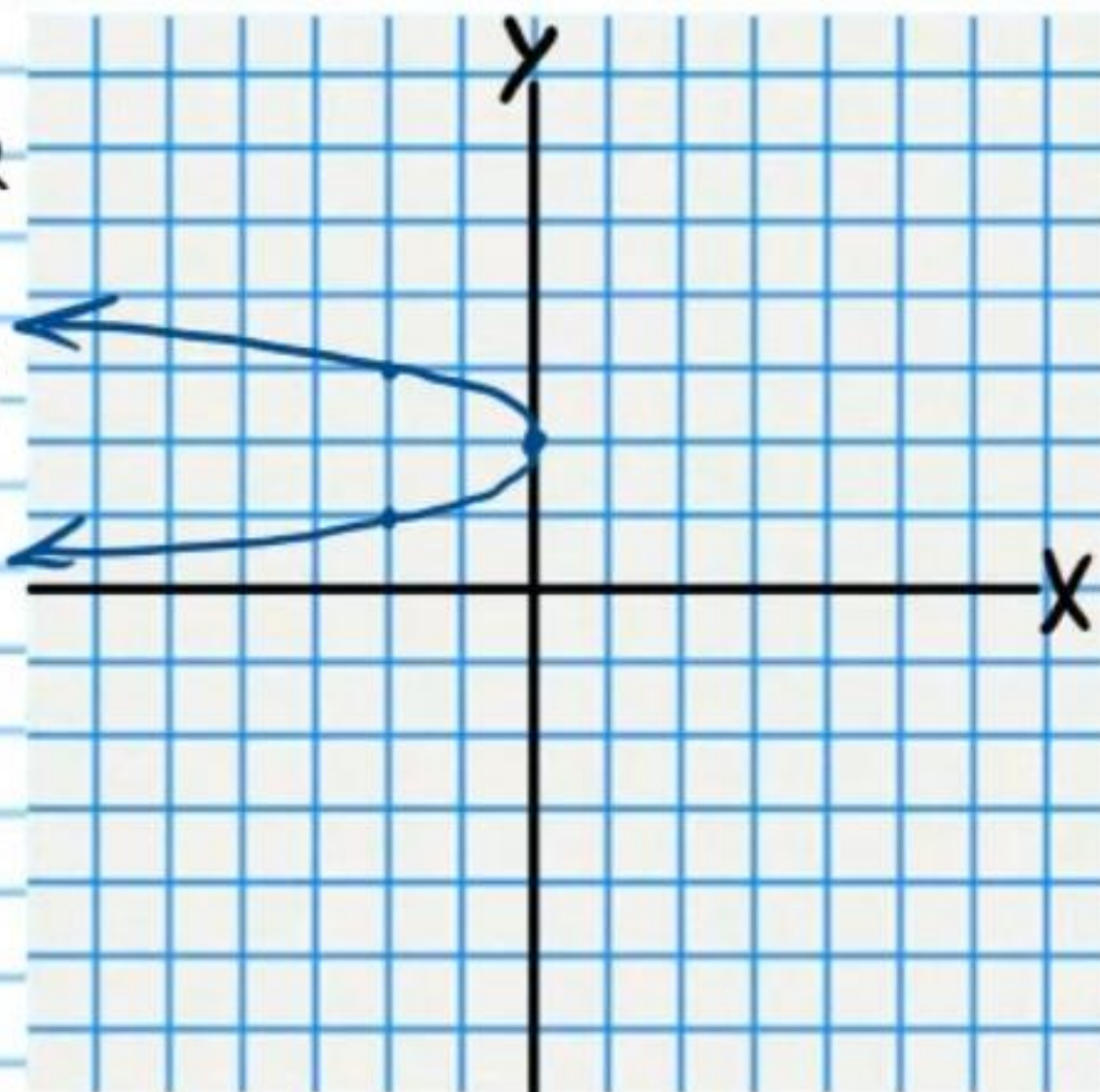
- (24) Vertex $(0, 2)$ and focus $(-\frac{1}{8}, 2)$.

$$P = \frac{1}{4a} \quad x - h = a(y - k)^2$$

$$\frac{1}{8} = \frac{1}{4a} \quad x - 0 = -2(y - 2)^2$$

$$a = 2$$

$$\boxed{x = -2(y - 2)^2}$$



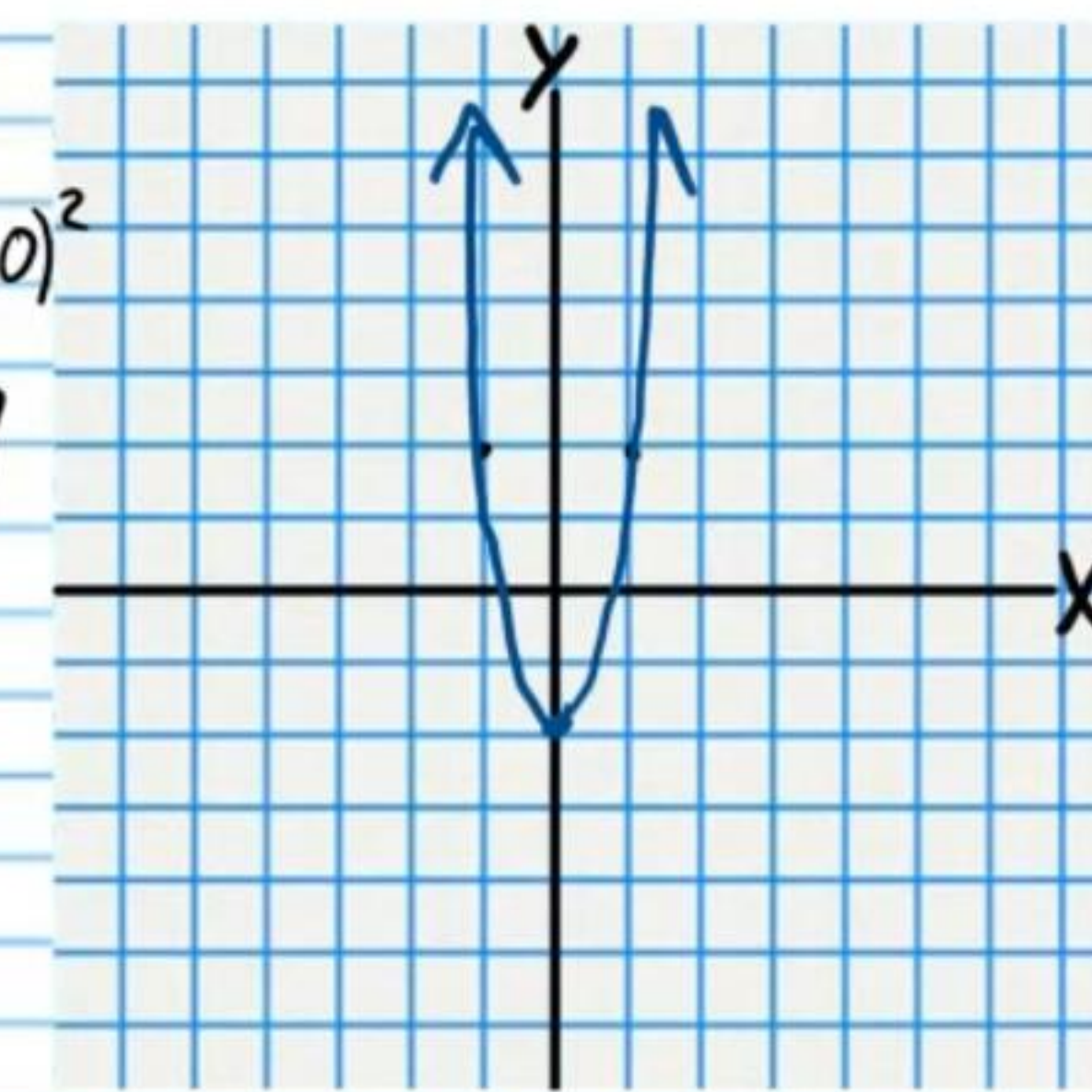
- (25) Vertex $(0, -1\frac{5}{16})$ and directrix $y = -2$.

$$P = \frac{1}{4a} \quad y - k = a(x - h)^2$$

$$\frac{1}{16} = \frac{1}{4a} \quad y + 1\frac{5}{16} = 4(x - 0)^2$$

$$a = 4$$

$$\boxed{y + 1\frac{5}{16} = 4x^2}$$



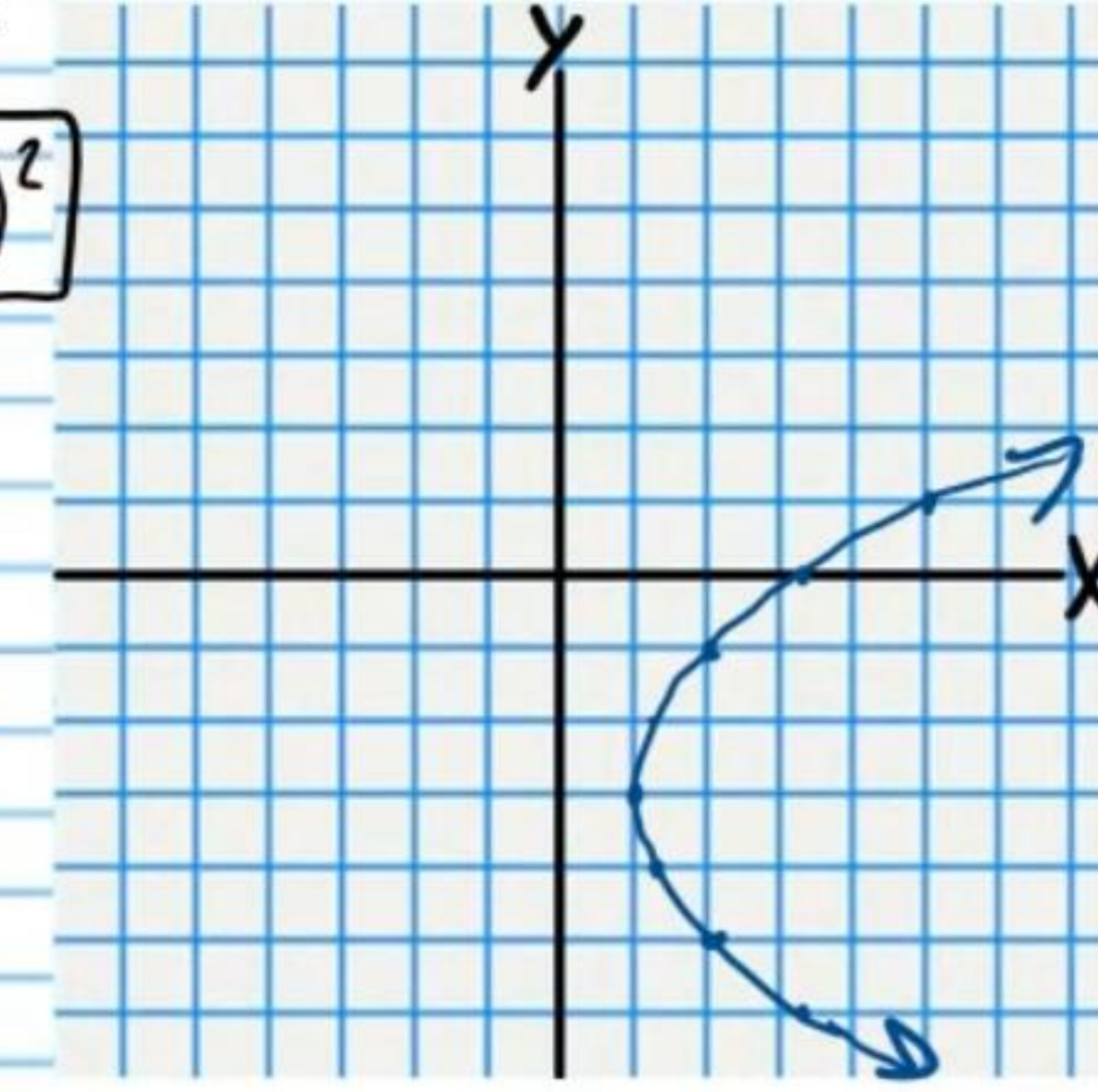
- (26) Focus $(2, -3)$ and directrix $x = 0$.

$$P = \frac{1}{4a} \quad x - h = a(y - k)^2$$

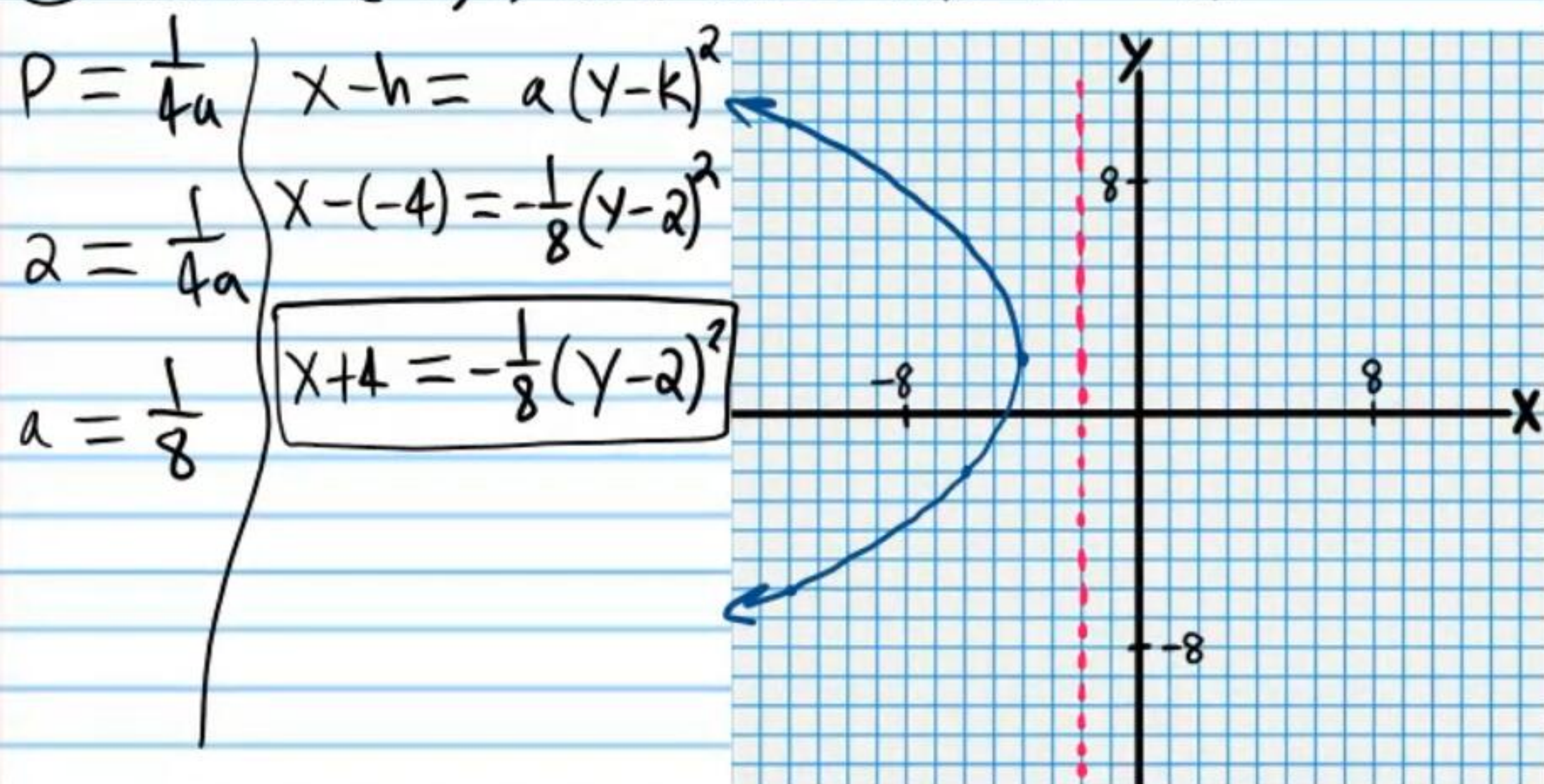
$$1 = \frac{1}{4a} \quad x - 1 = \frac{1}{4}(y + 3)^2$$

$$a = \frac{1}{4}$$

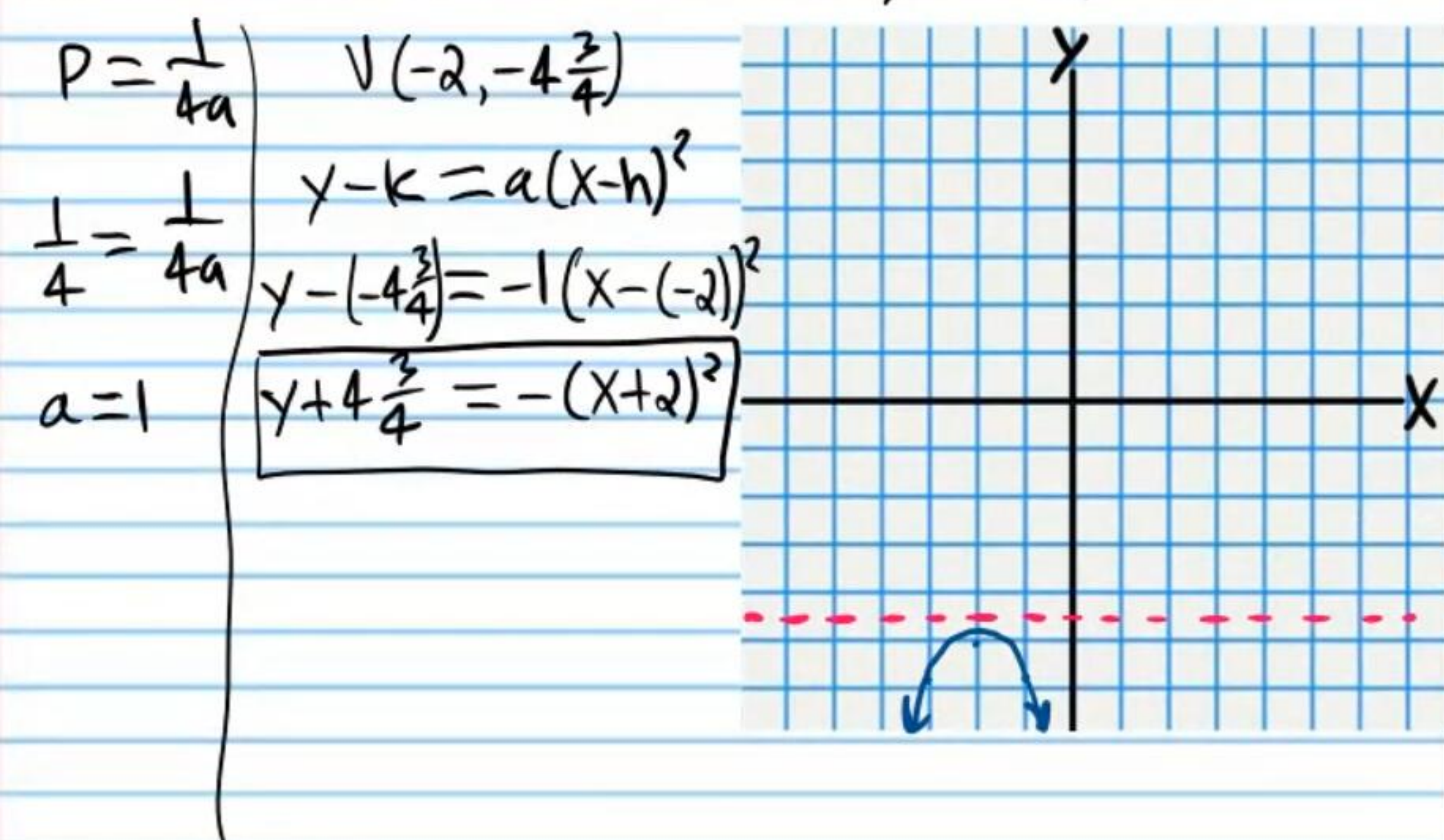
$$\boxed{x - 1 = \frac{1}{4}(y + 3)^2}$$



(27) Vertex $(-4, 2)$ and directrix $x = -2$.



(28) Focus $(-2, -5)$ and directrix $y = -4\frac{1}{2}$.

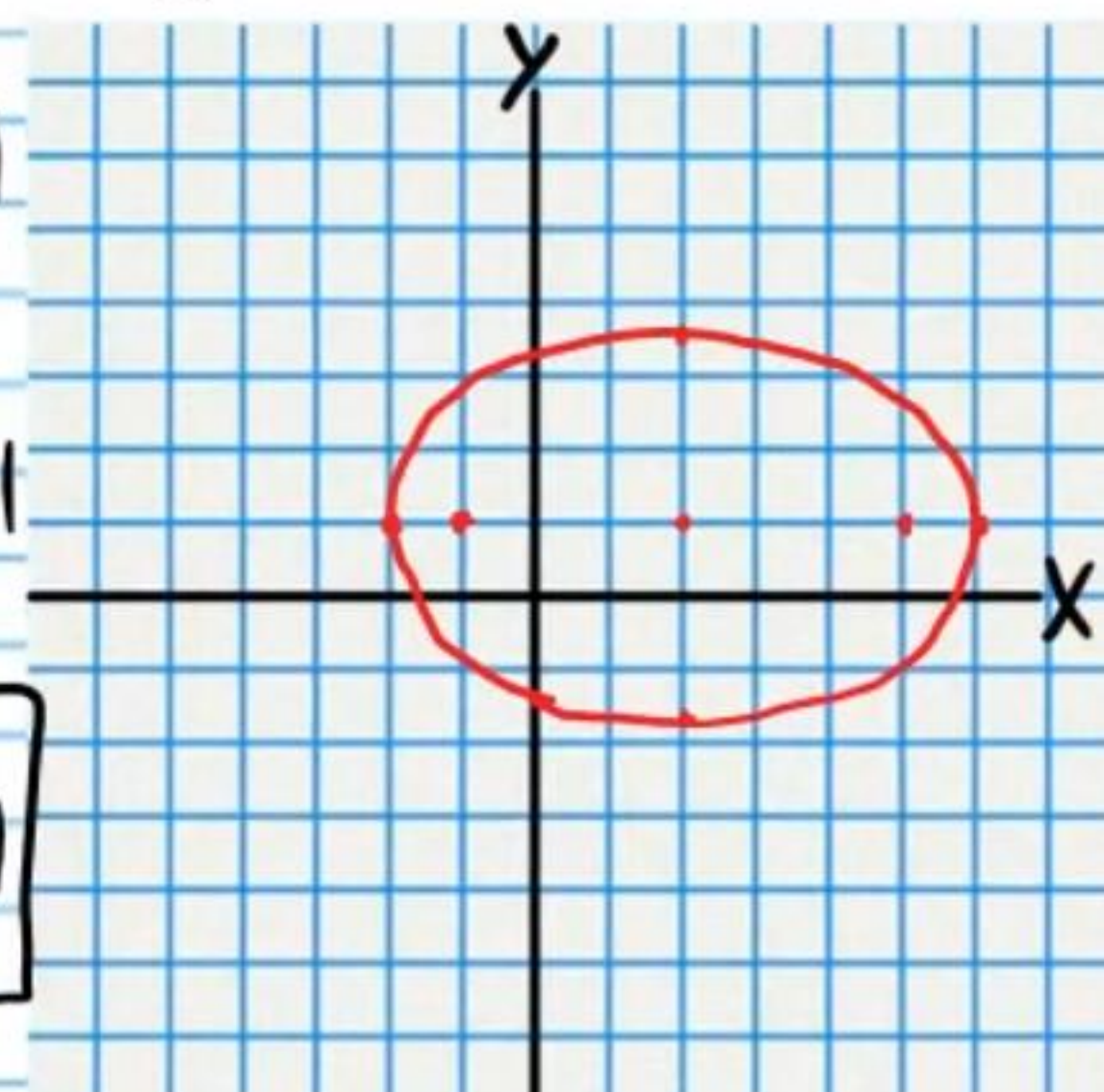


Find equations of ellipses with the following characteristics and graph each equation.

(29) Foci at $(5, 1)$ and $(-1, 1)$, length of major axis 8.

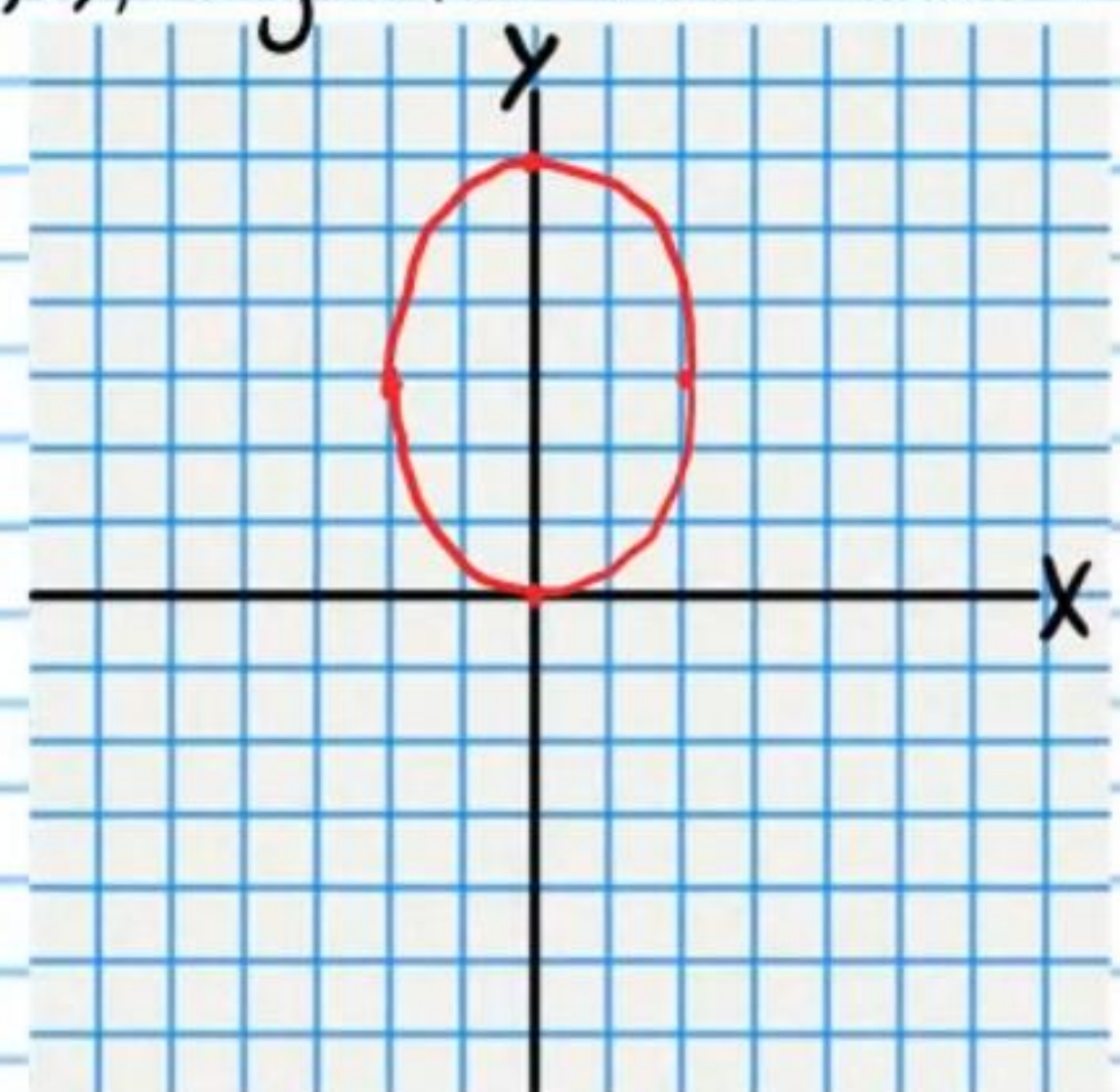
$a^2 = b^2 + c^2$
 $4^2 = b^2 + 3^2$
 $b = \sqrt{7}$

$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$
 $\frac{(x-2)^2}{4^2} + \frac{(y-1)^2}{(\sqrt{7})^2} = 1$
 $\frac{(x-2)^2}{16} + \frac{(y-1)^2}{7} = 1$



(30) Vertices at $(0, 6)$ and $(0, 0)$, length of minor axis 4.

$\frac{(y-k)^2}{a^2} + \frac{(x-h)^2}{b^2} = 1$
 $\frac{(y-3)^2}{3^2} + \frac{(x-0)^2}{2^2} = 1$
 $\frac{(y-3)^2}{9} + \frac{x^2}{4} = 1$



③ Foci at $(3, -5)$ and $(3, 5)$, length of major axis 14.

$$a^2 = b^2 + c^2$$

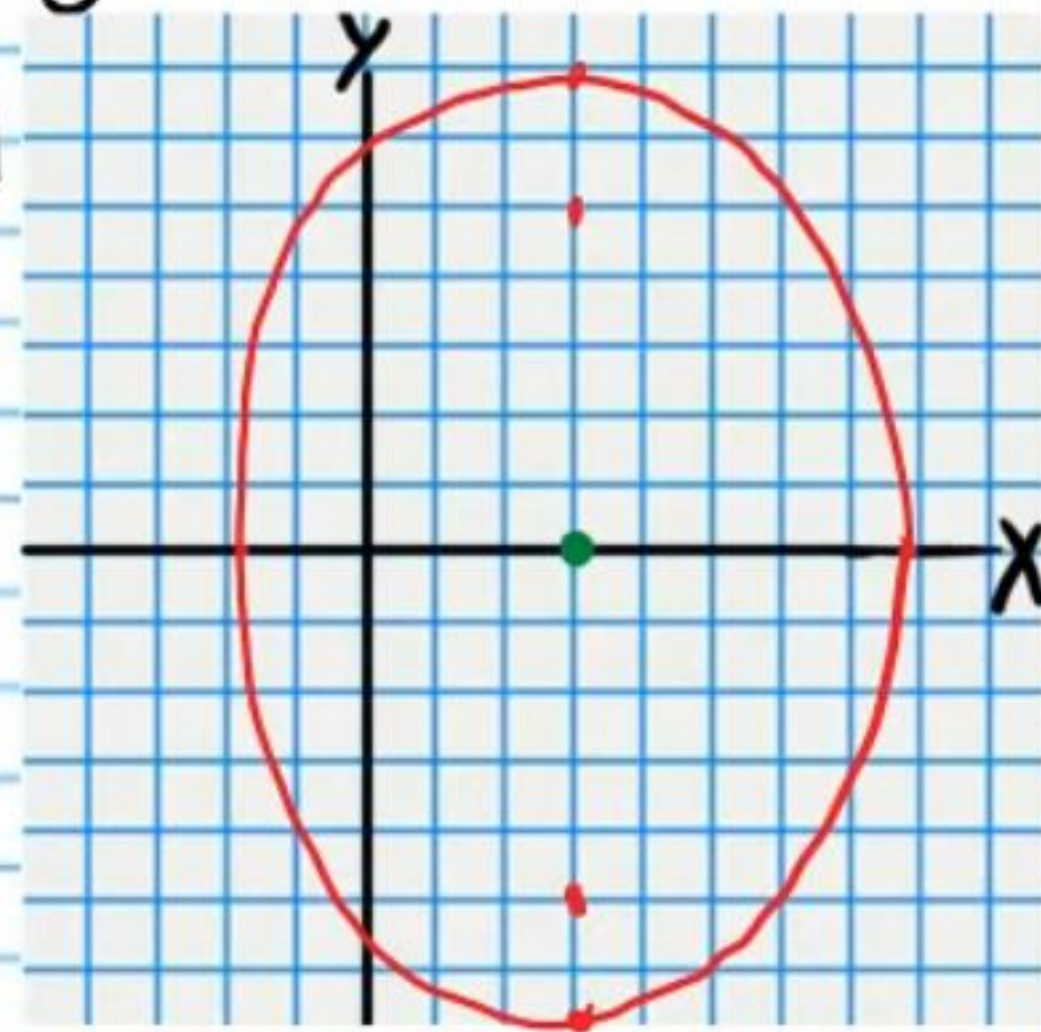
$$7^2 = b^2 + 5^2$$

$$b = \sqrt{24}$$

$$b = 2\sqrt{6}$$

$$\frac{(y-0)^2}{7^2} + \frac{(x-3)^2}{(\sqrt{24})^2} = 1$$

$$\boxed{\frac{y^2}{49} + \frac{(x-3)^2}{24} = 1}$$

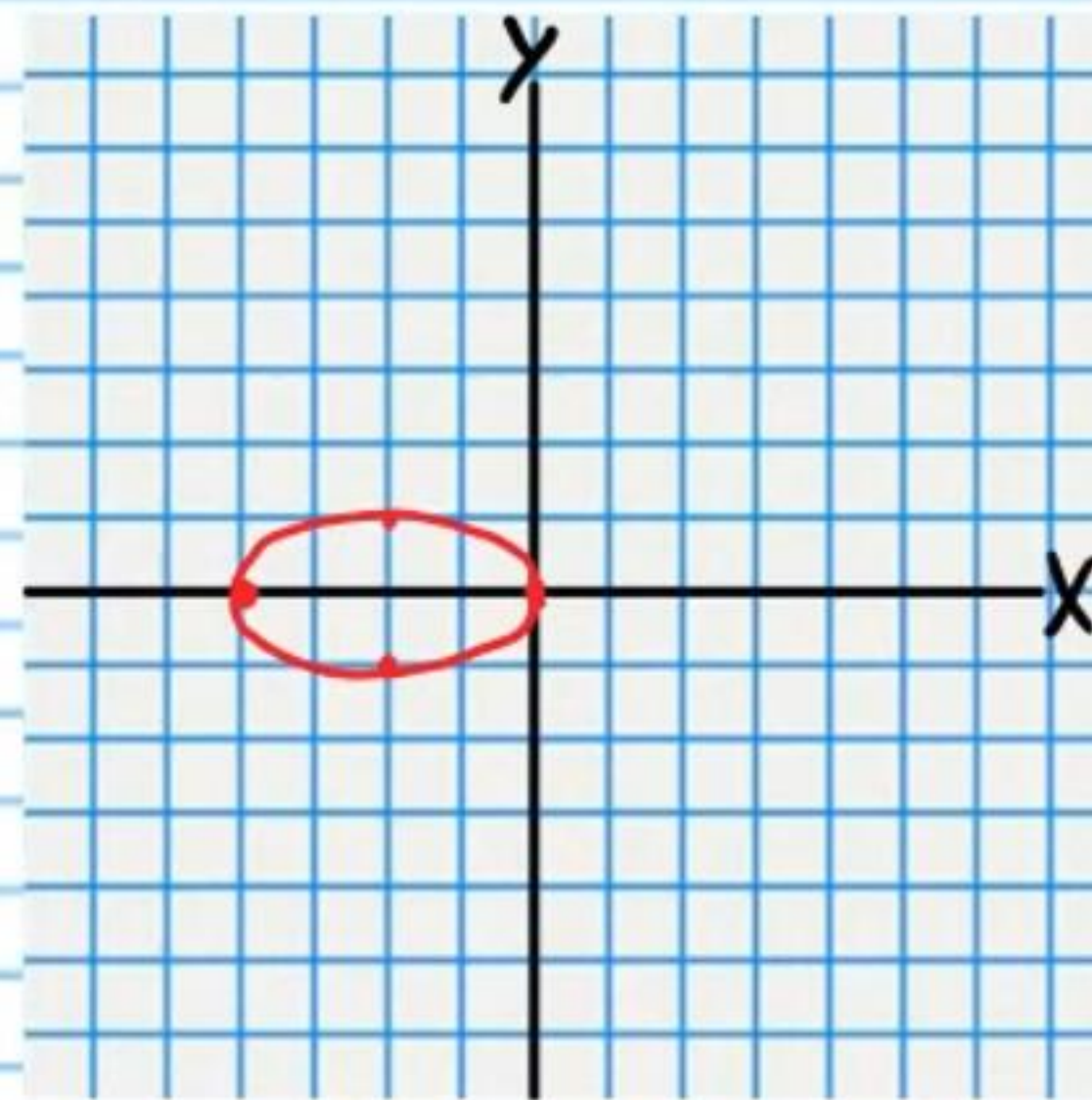


③ Vertices at $(-4, 0)$ and $(0, 0)$, length of minor axis 2.

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

$$\frac{(x+2)^2}{2^2} + \frac{(y-0)^2}{1^2} = 1$$

$$\boxed{\frac{(x+2)^2}{4} + y^2 = 1}$$

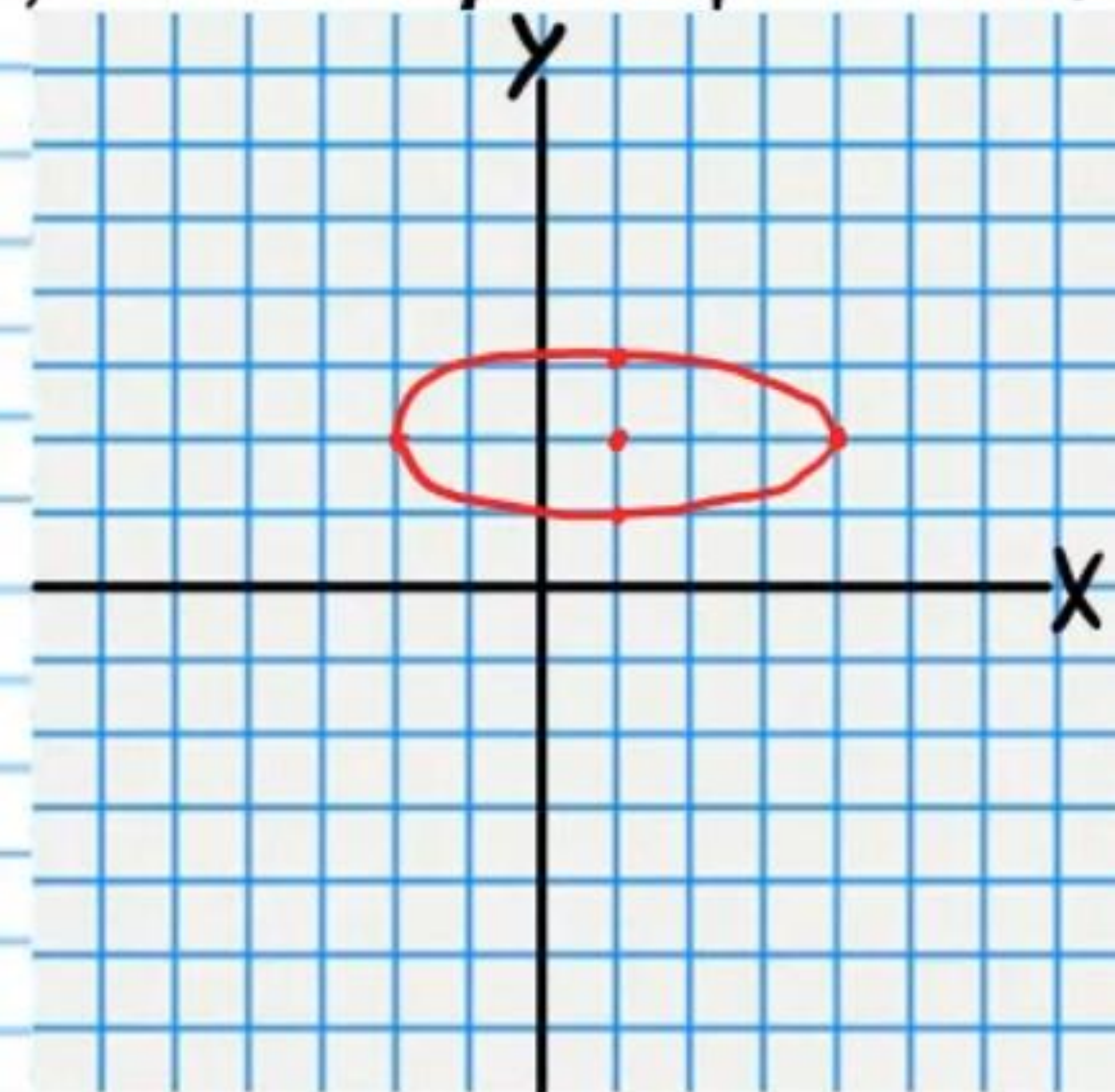


• ③ Center $(1, 2)$, Vertex $(4, 2)$, containing the point $(1, 3)$.

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

$$\frac{(x-1)^2}{3^2} + \frac{(y-2)^2}{1^2} = 1$$

$$\boxed{\frac{(x-1)^2}{9} + (y-2)^2 = 1}$$



③ Vertices $(4, 1)$ and $(4, 7)$, focus $(4, 6)$.

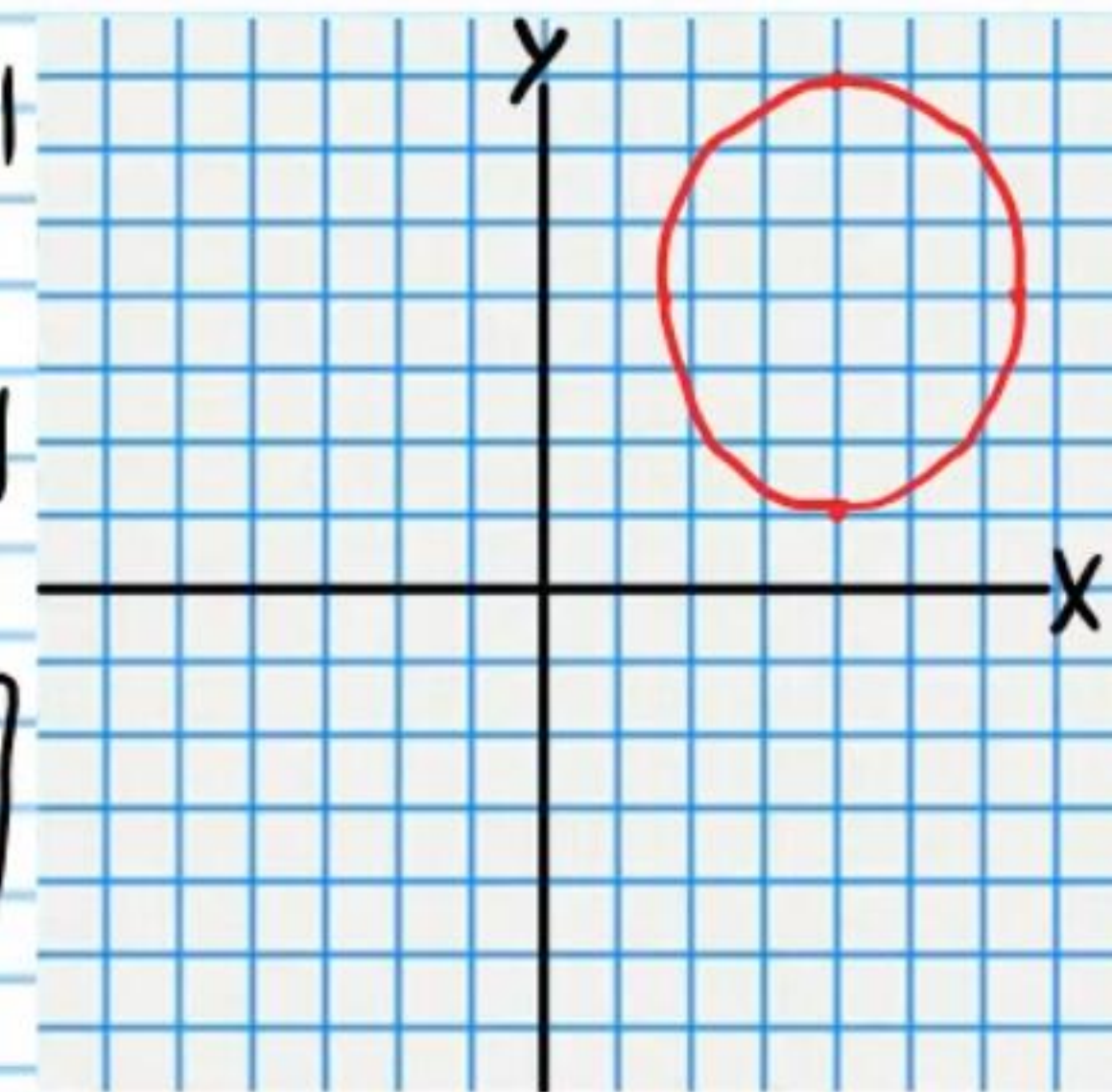
$$a^2 = b^2 + c^2$$

$$3^2 = b^2 + 2^2$$

$$b = \sqrt{5}$$

$$\frac{(y-4)^2}{3^2} + \frac{(x-4)^2}{(\sqrt{5})^2} = 1$$

$$\boxed{\frac{(y-4)^2}{9} + \frac{(x-4)^2}{5} = 1}$$



- (35) Center (1,2), focus (1,4), containing the point (2,2).

$$a^2 = b^2 + c^2$$

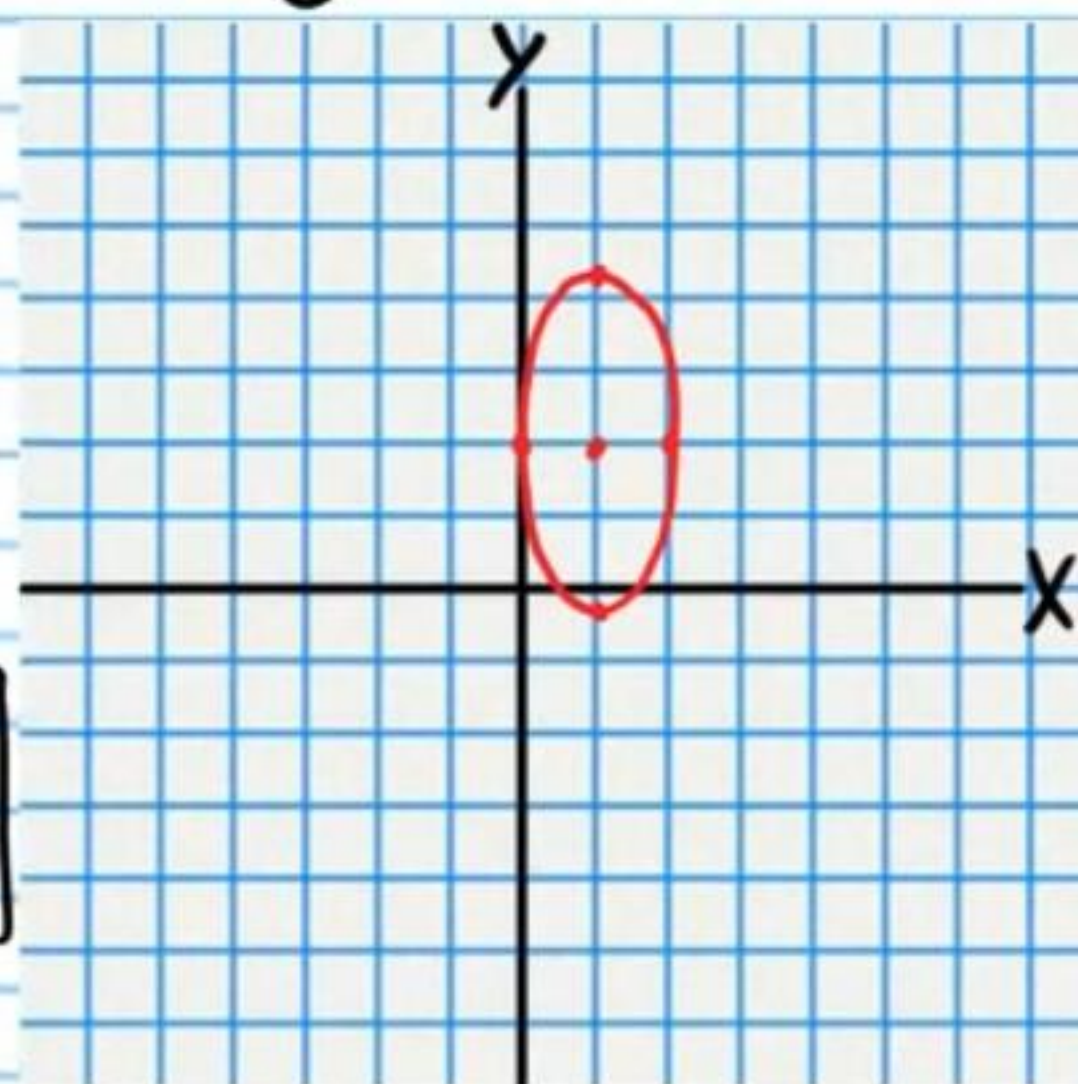
$$a^2 = 1^2 + 2^2$$

$$a = \sqrt{5}$$

$$\frac{(y-k)^2}{a^2} + \frac{(x-h)^2}{b^2} = 1$$

$$\frac{(y-2)^2}{(\sqrt{5})^2} + \frac{(x-1)^2}{1^2} = 1$$

$$\boxed{\frac{(y-2)^2}{5} + (x-1)^2 = 1}$$



- (36) Center at (2,-2), vertex at (7,-2), focus at (4,-2).

$$a^2 = b^2 + c^2$$

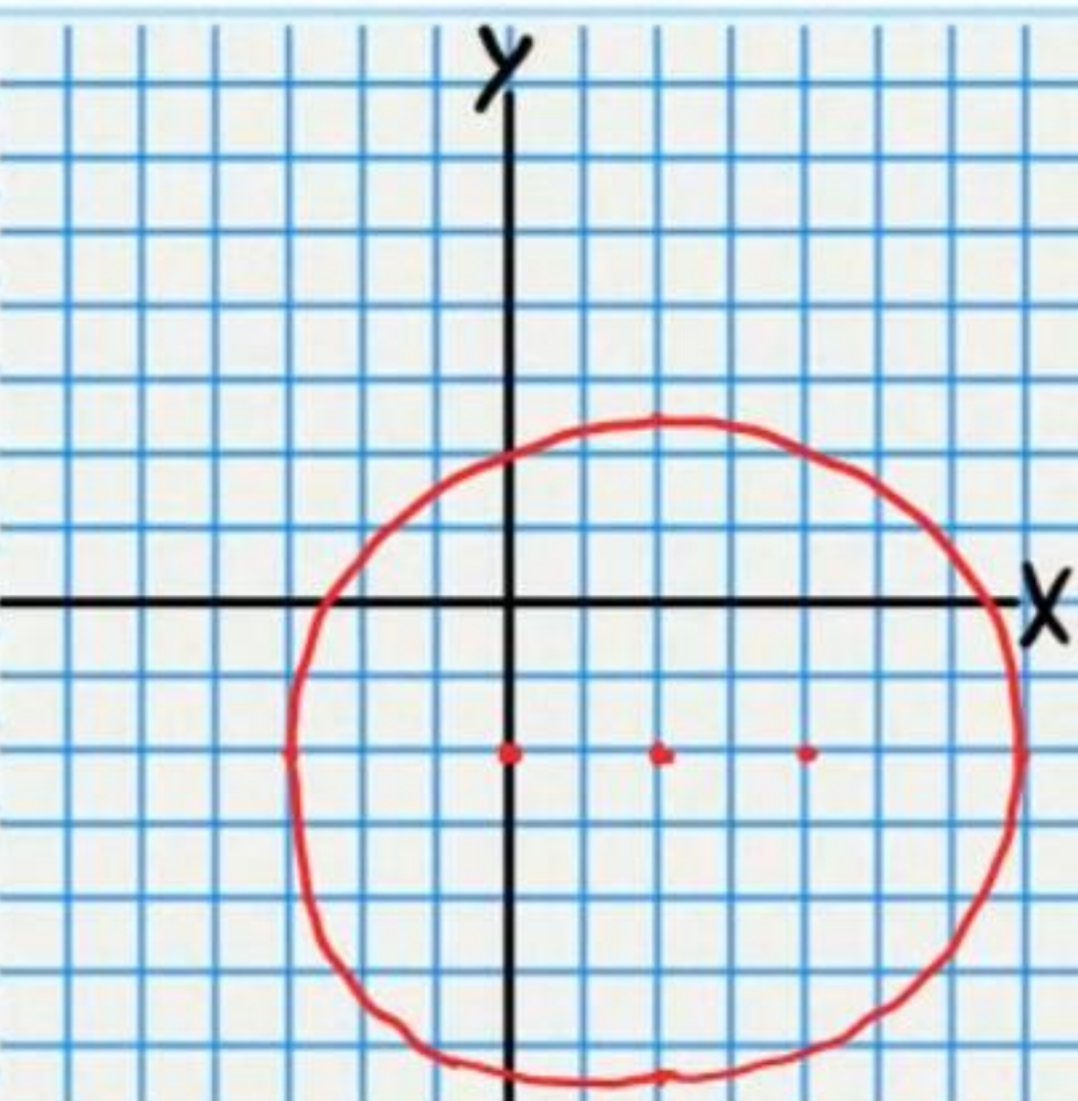
$$5^2 = b^2 + 2^2$$

$$b = \sqrt{21}$$

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

$$\frac{(x-2)^2}{5^2} + \frac{(y-(-2))^2}{(\sqrt{21})^2} = 1$$

$$\boxed{\frac{(x-2)^2}{25} + \frac{(y+2)^2}{21} = 1}$$



Find equations of hyperbolas with the following characteristics and graph each equation.

- (37) Center at (1,4), focus at (-2,4), vertex at (0,4).

$$a^2 + b^2 = c^2$$

$$1^2 + b^2 = 3^2$$

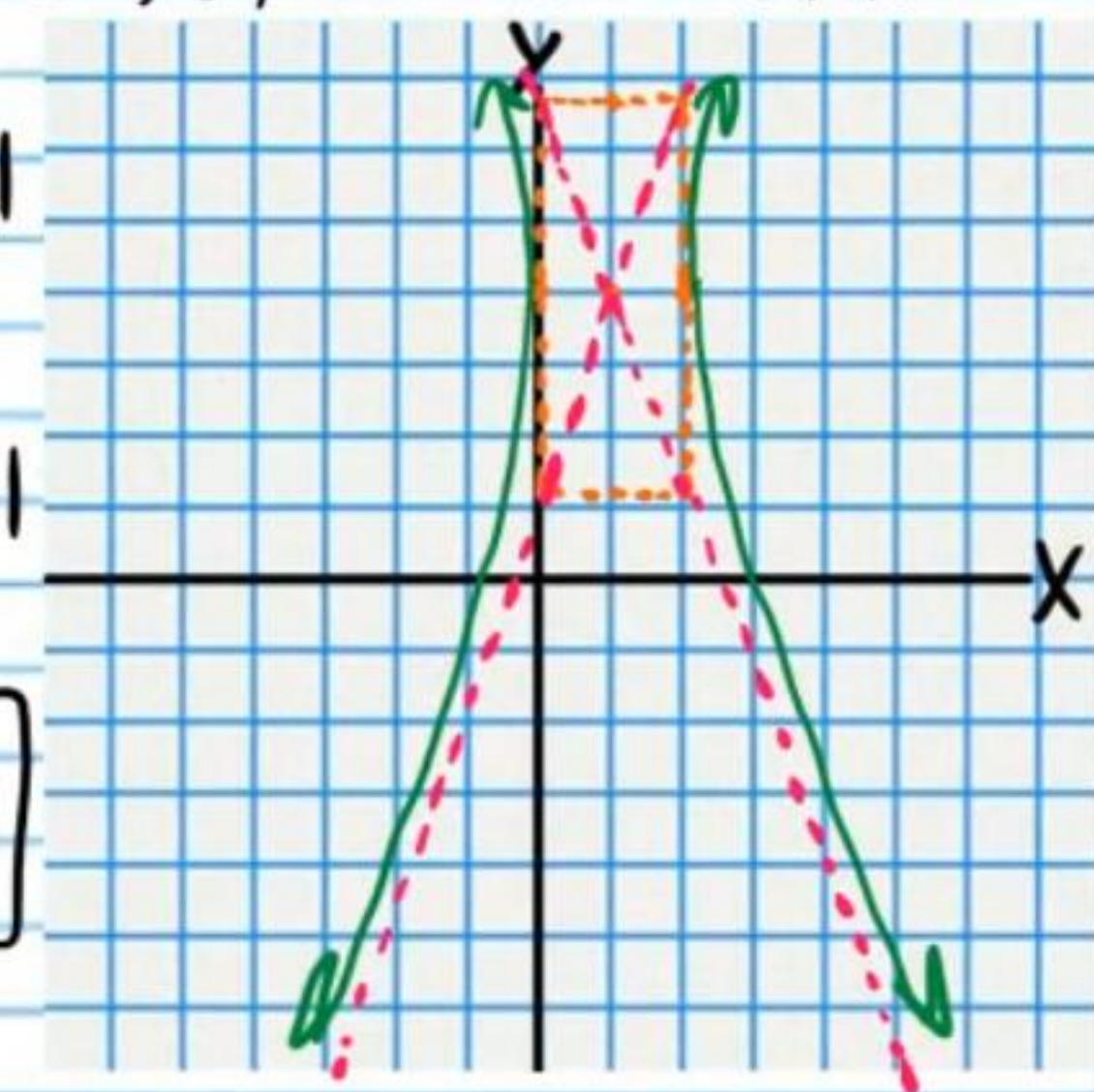
$$b = \sqrt{8}$$

$$b = 2\sqrt{2}$$

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

$$\frac{(x-1)^2}{1^2} - \frac{(y-4)^2}{(\sqrt{8})^2} = 1$$

$$\boxed{(x-1)^2 - \frac{(y-4)^2}{8} = 1}$$



- (38) Center at (-3,-2), focus at (-3,-6), vertex at (-3,0).

$$a^2 + b^2 = c^2$$

$$2^2 + b^2 = 4^2$$

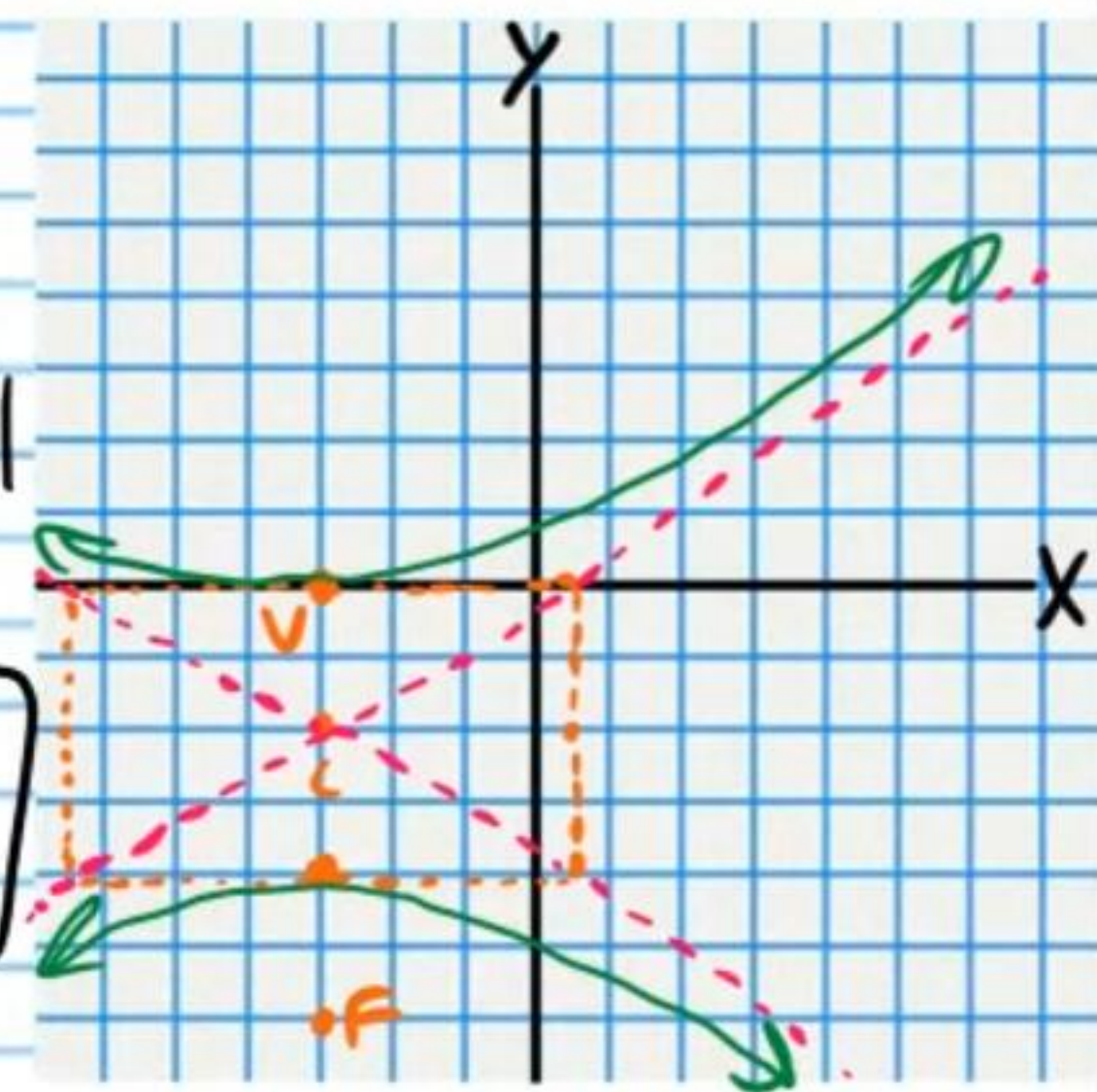
$$b = \sqrt{12}$$

$$b = 2\sqrt{3}$$

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

$$\frac{(y-(-2))^2}{2^2} - \frac{(x-(-3))^2}{(\sqrt{12})^2} = 1$$

$$\boxed{\frac{(y+2)^2}{4} - \frac{(x+3)^2}{12} = 1}$$



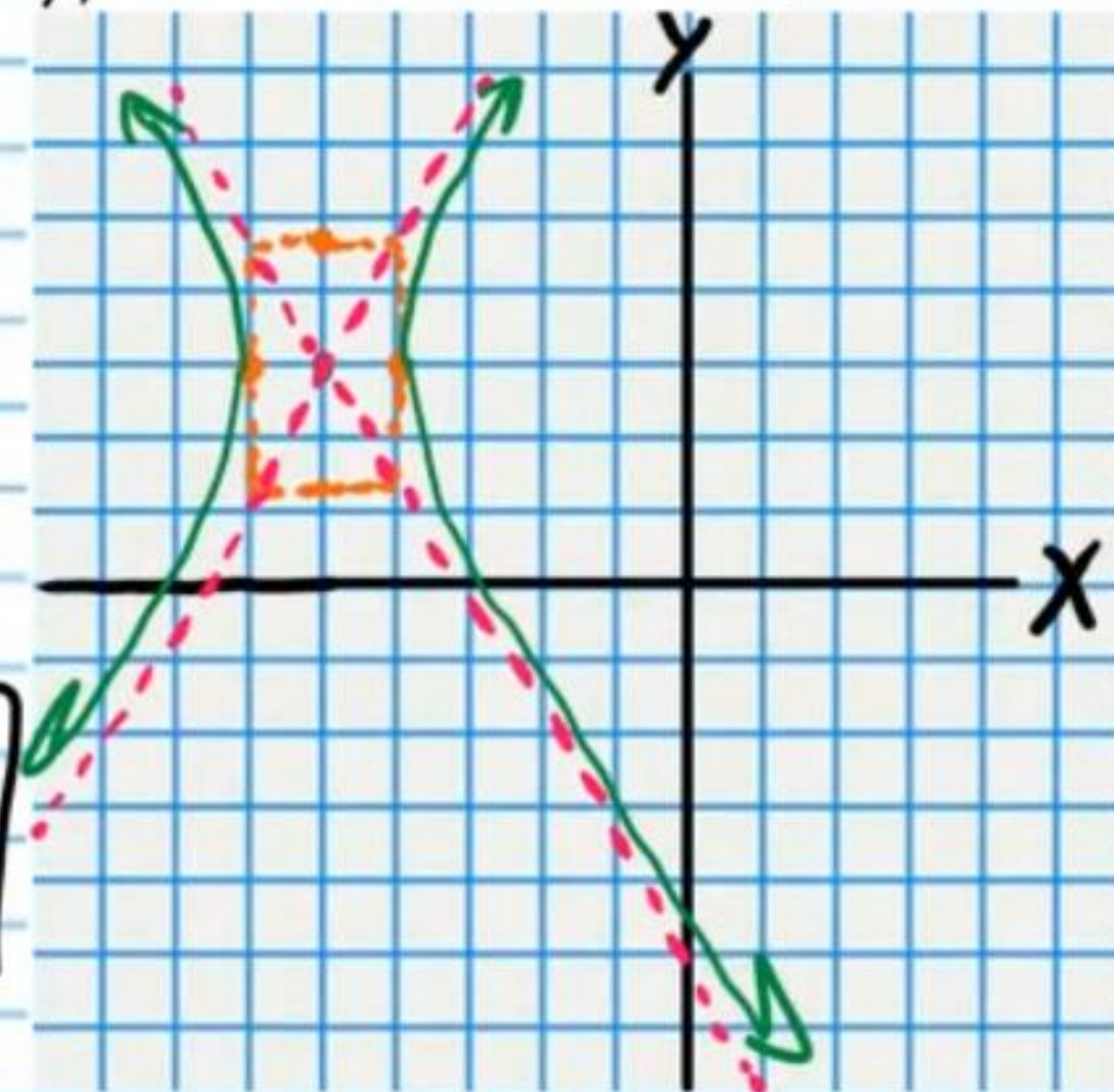
- ③ Foci at $(-3, 3)$ and $(-7, 3)$, Vertex at $(-6, 3)$

$$\begin{aligned} a^2 + b^2 &= c^2 \\ 1^2 + b^2 &= 2^2 \end{aligned}$$

$$b = \sqrt{3} \quad \frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

$$\frac{(x-(-5))^2}{1^2} - \frac{(y-3)^2}{(\sqrt{3})^2} = 1$$

$$\boxed{\frac{(x+5)^2}{1} - \frac{(y-3)^2}{3} = 1}$$



- ④ Focus at $(-4, 0)$, vertices at $(-4, 4)$ and $(-4, 2)$.

$$\begin{aligned} a^2 + b^2 &= c^2 \\ 1^2 + b^2 &= 3^2 \end{aligned}$$

$$b = \sqrt{8} \quad \frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

$$b = 2\sqrt{2} \quad \frac{(y-3)^2}{1^2} - \frac{(x-(-4))^2}{(\sqrt{8})^2} = 1$$

$$\boxed{\frac{(y-3)^2}{1} - \frac{(x+4)^2}{8} = 1}$$

